where

$$M = \frac{1}{\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + \left(2\zeta \frac{\omega}{\omega_n}\right)^2}}, \qquad \alpha = -\tan^{-1} \frac{2\zeta \frac{\omega}{\omega_n}}{1 - \frac{\omega^2}{\omega_n^2}}$$

As given by Equation (8–6), for $0 \le \zeta \le 0.707$ the maximum value of M occurs at the frequency ω_r, where

$$\omega_r = \omega_n \sqrt{1 - 2\zeta^2} = \omega_n \sqrt{\cos 2\theta} \tag{8–10}$$

The angle θ is defined in Figure 8–80. The frequency ω_r is the resonant frequency. At the resonant frequency, the value of M is maximum and is given by Equation (8–7), rewritten

$$M_r = \frac{1}{2\zeta\sqrt{1 - \zeta^2}} = \frac{1}{\sin 2\theta} \tag{8–11}$$

where M_r is defined as the *resonant peak magnitude.* The resonant peak magnitude is related to the damping of the system.

The magnitude of the resonant peak gives an indication of the relative stability of the system. A large resonant peak magnitude indicates the presence of a pair of dominant closed-loop poles with small damping ratio, which will yield an undesirable transient response. A smaller resonant peak magnitude, on the other hand, indicates the absence of a pair of dominant closed-loop poles with small damping ratio, meaning that the system is well damped.

Remember that ω_r is real only if $\zeta < 0.707$. Thus, there is no closed-loop resonance if $\zeta > 0.707$. [The value of M_r is unity only if $\zeta > 0.707$. See Equation (8–8).] Since the values of M_r and ω_r can be easily measured in a physical system, they are quite useful for checking agreement between theoretical and experimental analyses.

It is noted, however, that in practical design problems the phase margin and gain margin are more frequently specified than the resonant peak magnitude to indicate the degree of damping in a system.

Correlation between step transient response and frequency response in the standard second-order system. The maximum overshoot in the unit-step response of the standard second-order system, as shown in Figure 8–79, can be exactly correlated

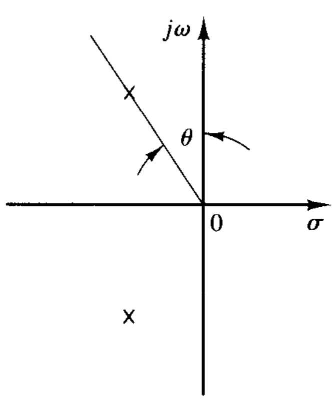

Figure 8–80
Definition of the angle θ.

to the resonant peak magnitude in the frequency response. Hence, essentially the same information of the system dynamics is contained in the frequency response as is in the transient response.

For a unit-step input, the output of the system shown in Figure 8–79 is given by Equation (4–21), or

$$c(t) = 1 - e^{-\zeta\omega_n t}\left(\cos \omega_d t + \frac{\zeta}{\sqrt{1 - \zeta^2}} \sin \omega_d t\right), \qquad \text{for } t \geq 0$$

where

$$\omega_d = \omega_n\sqrt{1 - \zeta^2} = \omega_n \cos\theta \tag{8–12}$$

On the other hand, the maximum overshoot M_p for the unit-step response is given by Equation (4–30), or

$$M_p = e^{-(\zeta/\sqrt{1-\zeta^2})\pi} \tag{8–13}$$

This maximum overshoot occurs in the transient response that has the damped natural frequency $\omega_d = \omega_n\sqrt{1 - \zeta^2}$. The maximum overshoot becomes excessive for values of $\zeta < 0.4$.

Since the second-order system shown in Figure 8–79 has the open-loop transfer function

$$G(s) = \frac{\omega_n^2}{s(s + 2\zeta\omega_n)}$$

for sinusoidal operation, the magnitude of $G(j\omega)$ becomes unity when

$$\omega = \omega_n\sqrt{\sqrt{1 + 4\zeta^4} - 2\zeta^2}$$

which can be obtained by equating $|G(j\omega)|$ to unity and solving for ω. At this frequency, the phase angle of $G(j\omega)$ is

$$\underline{/G(j\omega)} = -\underline{/j\omega} - \underline{/j\omega + 2\zeta\omega_n} = -90^\circ - \tan^{-1}\frac{\sqrt{\sqrt{1 + 4\zeta^4} - 2\zeta^2}}{2\zeta}$$

Thus, the phase margin γ is

$$\begin{aligned}\gamma &= 180^\circ + \underline{/G(j\omega)} \\ &= 90^\circ - \tan^{-1}\frac{\sqrt{\sqrt{1 + 4\zeta^4} - 2\zeta^2}}{2\zeta} \\ &= \tan^{-1}\frac{2\zeta}{\sqrt{\sqrt{1 + 4\zeta^4} - 2\zeta^2}}\end{aligned} \tag{8–14}$$

Equation (8–14) gives the relationship between the damping ratio ζ and the phase margin γ. (Notice that the phase margin γ is a function only of the damping ratio ζ.)

In the following, we shall summarize the correlation between the step transient response and frequency response of the second-order system given by Equation (8–9):

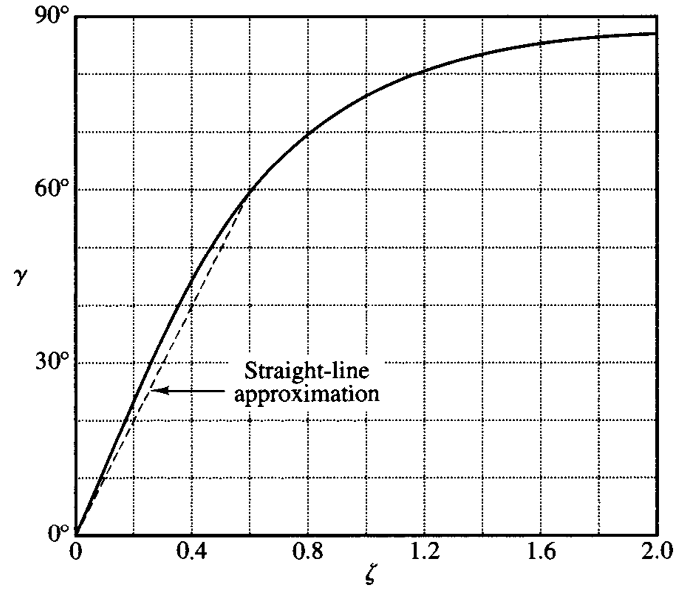

Figure 8–81
Curve γ (phase margin) versus ζ for the system shown in Figure 8–79.

1. The phase margin and the damping ratio are directly related. Figure 8–81 shows a plot of the phase margin γ as a function of the damping ratio ζ. It is noted that for the standard second-order system shown in Figure 8–79 the phase margin γ and the damping ratio ζ are related approximately by a straight line for $0 \leq \zeta \leq 0.6$, as follows:

$$\zeta = \frac{\gamma}{100}$$

Thus a phase margin of 60° corresponds to a damping ratio of 0.6. For higher-order systems having a dominant pair of closed-loop poles, this relationship may be used as a rule of thumb in estimating the relative stability in transient response (that is, the damping ratio) from the frequency response.

2. Referring to Equations (8–10) and (8–12), we see that the values of ω_r and ω_d are almost the same for small values of ζ. Thus, for small values of ζ, the value of ω_r is indicative of the speed of the transient response of the system.

3. From Equations (8–11) and (8–13), we note that the smaller the value of ζ is, the larger the values of M_r and M_p are. The correlation between M_r and M_p as a function of ζ is shown in Figure 8–82. A close relationship between M_r and M_p can be seen for $\zeta > 0.4$. For very small values of ζ, M_r becomes very large ($M_r \gg 1$), while the value of M_p does not exceed 1.

Correlation between step transient response and frequency response in general systems. The design of control systems is very often carried out on the basis of frequency response. The main reason for this is the relative simplicity of this approach as compared to others. Since in many applications it is the transient response of the

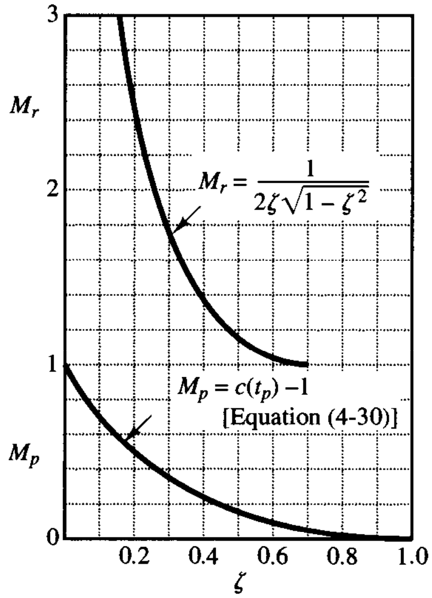

Figure 8–82
Curves M_r versus ζ and M_p versus ζ for the system shown in Figure 8–79.

system to aperiodic inputs rather than the steady-state response to sinusoidal inputs that is of primary concern, the question of correlation between transient response and frequency response arises.

For the second-order system shown in Figure 8–79, mathematical relationships correlating the step transient response and frequency response can be obtained easily. The time response of a second-order system can be exactly predicted from a knowledge of the M_r and ω_r of its closed-loop frequency response.

For higher-order systems, the correlation is more complex, and transient response may not be predicted easily from frequency response because additional poles may change the correlation between the step transient response and frequency response existing for a second-order system. Mathematical techniques for obtaining the exact correlation are available, but they are very laborious and of little practical value.

The applicability of the transient-response–frequency-response correlation existing for the second-order system shown in Figure 8–79 to higher-order systems depends on the presence of a dominant pair of complex-conjugate closed-loop poles in the latter systems. Clearly, if the frequency response of a higher-order system is dominated by a pair of complex-conjugate closed-loop poles, the transient-response–frequency-response correlation existing for the second-order system can be extended to the higher-order system.

For linear, time-invariant, higher-order systems having a dominant pair of complex-conjugate closed-loop poles, the following relationships generally exist between the step transient response and frequency response:

1. The value of M_r is indicative of the relative stability. Satisfactory transient performance is usually obtained if the value of M_r is in the range $1.0 < M_r < 1.4$ (0 dB <

$M_r < 3$ dB), which corresponds to an effective damping ratio of $0.4 < \zeta < 0.7$. For values of M_r greater than 1.5, the step transient response may exhibit several overshoots. (Note that in general a large value of M_r corresponds to a large overshoot in the step transient response. If the system is subjected to noise signals whose frequencies are near the resonant frequency ω_r, the noise will be amplified in the output and will present serious problems.)

2. The magnitude of the resonant frequency ω_r is indicative of the speed of the transient response. The larger the value of ω_r, the faster the time response is. In other words, the rise time varies inversely with ω_r. In terms of the open-loop frequency response, the damped natural frequency in the transient response is somewhere between the gain crossover frequency and phase crossover frequency.

3. The resonant peak frequency ω_r and the damped natural frequency ω_d for step transient response are very close to each other for lightly damped systems.

The three relationships just listed are useful for correlating the step transient response with the frequency response of higher-order systems, provided they can be approximated by a second-order system or a pair of complex-conjugate closed-loop poles. If a higher-order system satisfies this condition, a set of time-domain specifications may be translated into frequency-domain specifications. This simplifies greatly the design work or compensation work of higher-order systems.

In addition to the phase margin, gain margin, resonant peak M_r, and resonant peak frequency ω_r, there are other frequency-domain quantities commonly used in performance specifications. They are the cutoff frequency, bandwidth, and the cutoff rate. These will be defined in what follows.

Cutoff frequency and bandwidth. Referring to Figure 8–83, the frequency ω_b at which the magnitude of the closed-loop frequency response is 3 dB below its zero-frequency value is called the *cutoff frequency*. Thus

$$\left|\frac{C(j\omega)}{R(j\omega)}\right| < \left|\frac{C(j0)}{R(j0)}\right| - 3\text{ dB}, \qquad \text{for } \omega > \omega_b$$

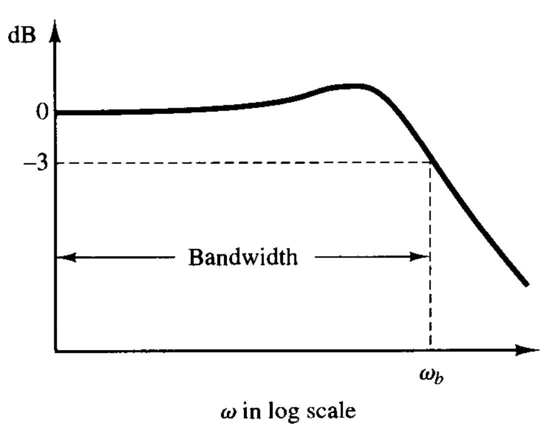

Figure 8–83
Plot of a closed-loop frequency response curve showing cutoff frequency ω_b and bandwidth.

For systems in which $|C(j0)/R(j0)| = 0$ dB,

$$\left|\frac{C(j\omega)}{R(j\omega)}\right| < -3 \text{ dB}, \qquad \text{for } \omega > \omega_b$$

The closed-loop system filters out the signal components whose frequencies are greater than the cutoff frequency and transmits those signal components with frequencies lower than the cutoff frequency.

The frequency range $0 \le \omega \le \omega_b$ in which the magnitude of the closed loop does not drop -3 dB is called the *bandwidth* of the system. The bandwidth indicates the frequency where the gain starts to fall off from its low-frequency value. Thus, the bandwidth indicates how well the system will track an input sinusoid. Note that for a given ω_n the rise time increases with increasing damping ratio ζ. On the other hand, the bandwidth decreases with the increase of ζ. Therefore, the rise time and the bandwidth are inversely proportional to each other.

The specification of the bandwidth may be determined by the following factors:

1. The ability to reproduce the input signal. A large bandwidth corresponds to a small rise time, or fast response. Roughly speaking, we can say that the bandwidth is proportional to the speed of response.
2. The necessary filtering characteristics for high-frequency noise.

For the system to follow arbitrary inputs accurately, it is necessary that the system have a large bandwidth. From the viewpoint of noise, however, the bandwidth should not be too large. Thus, there are conflicting requirements on the bandwidth, and a compromise is usually necessary for good design. Note that a system with large bandwidth requires high-performance components. So the cost of components usually increases with the bandwidth.

Cutoff rate. The cutoff rate is the slope of the log-magnitude curve near the cutoff frequency. The cutoff rate indicates the ability of a system to distinguish the signal from noise.

It is noted that a closed-loop frequency response curve with a steep cutoff characteristic may have a large resonant peak magnitude, which implies that the system has relatively small stability margin.

EXAMPLE 8–21 Consider the following two systems:

$$\text{System I: } \frac{C(s)}{R(s)} = \frac{1}{s+1}, \qquad \text{System II: } \frac{C(s)}{R(s)} = \frac{1}{3s+1}$$

Compare the bandwidths of these two systems. Show that the system with the larger bandwidth has a faster speed of response and can follow the input much better than the one with a smaller bandwidth.

Figure 8–84(a) shows the closed-loop frequency-response curves for the two systems. (Asymptotic curves are shown by dashed lines.) We find that the bandwidth of system I is $0 \le \omega \le 1$ rad/sec and that of system II is $0 \le \omega \le 0.33$ rad/sec. Figures 8–84(b) and (c) show, respectively, the unit-step response and unit-ramp response curves for the two systems. Clearly, system I, whose bandwidth is three times wider than that of system II, has a faster speed of response and can follow the input much better.

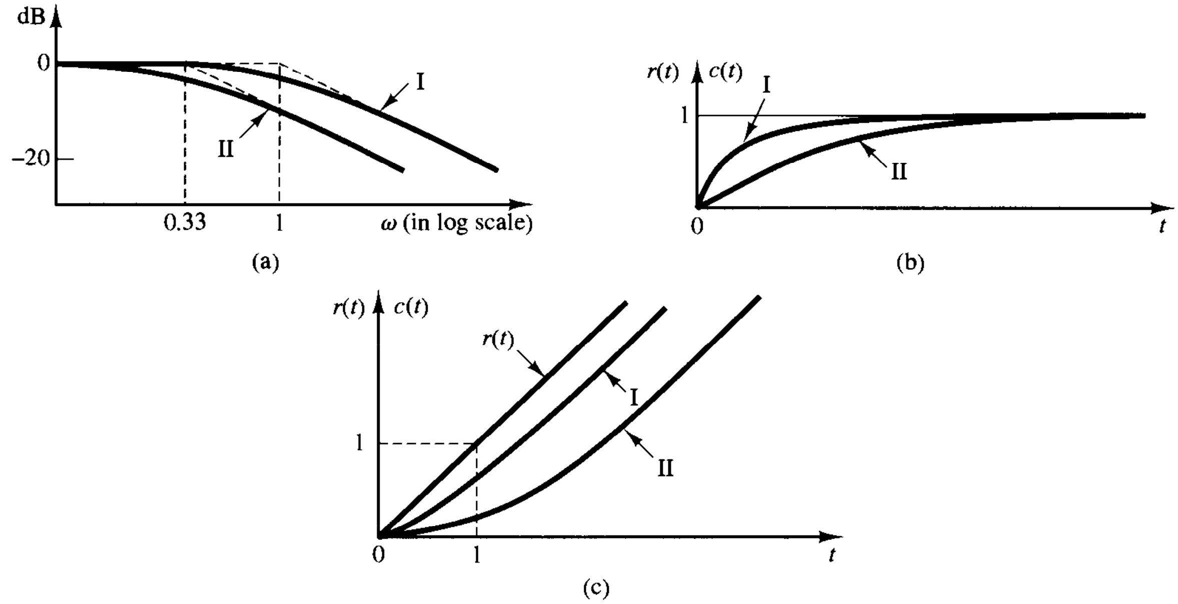

Figure 8–84 Comparison of dynamic characteristics of the two systems considered in Example 8–21. (a) Closed-loop frequency-response curves; (b) unit-step response curves; (c) unit-ramp response curves.

8–10 CLOSED-LOOP FREQUENCY RESPONSE

Closed-loop frequency response of unity-feedback systems. For a stable closed-loop system, the frequency response can be obtained easily from that of the open loop. Consider the unity-feedback system shown in Figure 8–85(a). The closed-loop transfer function is

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$$

In the Nyquist or polar plot shown in Figure 8–85(b), the vector $\overrightarrow{OA}$ represents $G(j\omega_1)$, where ω_1 is the frequency at point A. The length of the vector $\overrightarrow{OA}$ is $|G(j\omega_1)|$ and the angle of the vector $\overrightarrow{OA}$ is $\underline{/G(j\omega_1)}$. The vector $\overrightarrow{PA}$, the vector from the $-1 + j0$ point to the Nyquist locus, represents $1 + G(j\omega_1)$. Therefore, the ratio of $\overrightarrow{OA}$ to $\overrightarrow{PA}$ represents the closed-loop frequency response, or

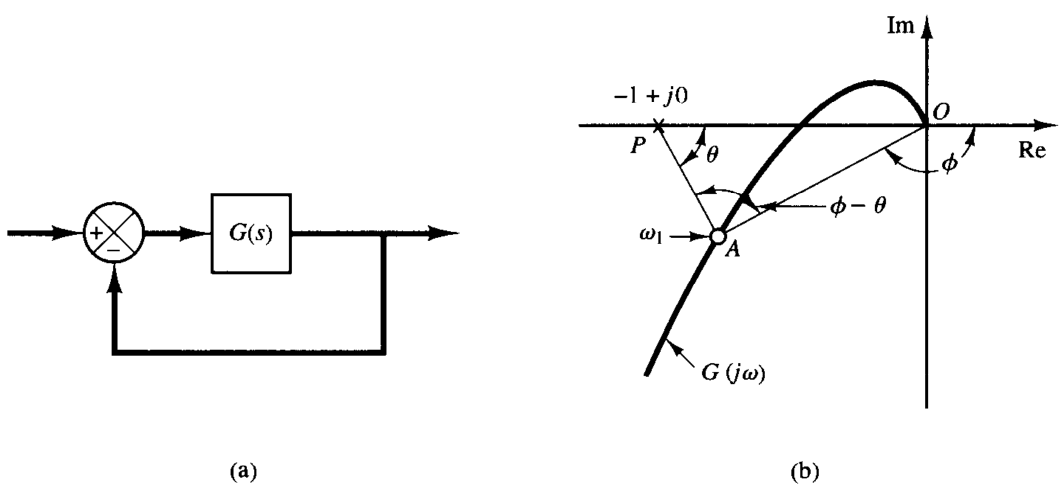

Figure 8–85 (a) Unity-feedback system; (b) determination of closed-loop frequency response from open-loop frequency response.

$$\frac{\overrightarrow{OA}}{\overrightarrow{PA}} = \frac{G(j\omega_1)}{1 + G(j\omega_1)} = \frac{C(j\omega_1)}{R(j\omega_1)}$$

The magnitude of the closed-loop transfer function at $\omega = \omega_1$ is the ratio of the magnitudes of $\overrightarrow{OA}$ to $\overrightarrow{PA}$. The phase angle of the closed-loop transfer function at $\omega = \omega_1$ is the angle formed by the vectors $\overrightarrow{OA}$ to $\overrightarrow{PA}$, that is $\phi - \theta$, shown in Figure 8–85(b). By measuring the magnitude and phase angle at different frequency points, the closed-loop frequency-response curve can be obtained.

Let us define the magnitude of the closed-loop frequency response as M and the phase angle as α, or

$$\frac{C(j\omega)}{R(j\omega)} = Me^{j\alpha}$$

In the following, we shall find the constant magnitude loci and constant phase-angle loci. Such loci are convenient in determining the closed-loop frequency response from the polar plot or Nyquist plot.

Constant-magnitude loci (M circles). To obtain the constant-magnitude loci, let us first note that $G(j\omega)$ is a complex quantity and can be written as follows:

$$G(j\omega) = X + jY$$

where X and Y are real quantities. Then M is given by

$$M = \frac{|X + jY|}{|1 + X + jY|}$$

and M^2 is

$$M^2 = \frac{X^2 + Y^2}{(1 + X)^2 + Y^2}$$

Hence

$$X^2(1 - M^2) - 2M^2X - M^2 + (1 - M^2)Y^2 = 0 \qquad (8\text{–}15)$$

If $M = 1$, then from Equation (8–15) we obtain $X = -\frac{1}{2}$. This is the equation of a straight line parallel to the Y axis and passing through the point $(-\frac{1}{2}, 0)$.

If $M \neq 1$, Equation (8–15) can be written

$$X^2 + \frac{2M^2}{M^2 - 1}X + \frac{M^2}{M^2 - 1} + Y^2 = 0$$

If the term $M^2/(M^2 - 1)^2$ is added to both sides of this last equation, we obtain

$$\left(X + \frac{M^2}{M^2 - 1}\right)^2 + Y^2 = \frac{M^2}{(M^2 - 1)^2} \qquad (8\text{–}16)$$

Equation (8–16) is the equation of a circle with center at $X = -M^2/(M^2 - 1)$, $Y = 0$ and with radius $|M/(M^2 - 1)|$.

The constant M loci on the $G(s)$ plane are thus a family of circles. The center and radius of the circle for a given value of M can be easily calculated. For example, for $M = 1.3$, the center is at $(-2.45, 0)$ and the radius is 1.88. A family of constant M circles is shown in Figure 8–86. It is seen that as M becomes larger compared with 1, the M circles become smaller and converge to the $-1 + j0$ point. For $M > 1$, the centers of the M circles lie to the left of the $-1 + j0$ point. Similarly, as M becomes smaller compared with 1, the M circle becomes smaller and converges to the origin. For $0 < M < 1$, the centers of the M circles lie to the right of the origin. $M = 1$ corresponds to the locus of points equidistant from the origin and from the $-1 + j0$ point. As stated earlier, it is a straight line passing through the point $(-\frac{1}{2}, 0)$ and parallel to the imaginary axis. (The constant M circles corresponding to $M > 1$ lie to the left of the $M = 1$ line and those corresponding to $0 < M < 1$ lie to the right of the $M = 1$ line.) The M circles are symmetrical with respect to the straight line corresponding to $M = 1$ and with respect to the real axis.

Constant phase-angle loci (N circles). We shall obtain the phase angle α in terms of X and Y. Since

$$\angle e^{j\alpha} = \angle \frac{X + jY}{1 + X + jY}$$

the phase angle α is

$$\alpha = \tan^{-1}\left(\frac{Y}{X}\right) - \tan^{-1}\left(\frac{Y}{1 + X}\right)$$

If we define

$$\tan \alpha = N$$

then

$$N = \tan\left[\tan^{-1}\left(\frac{Y}{X}\right) - \tan^{-1}\left(\frac{Y}{1 + X}\right)\right]$$

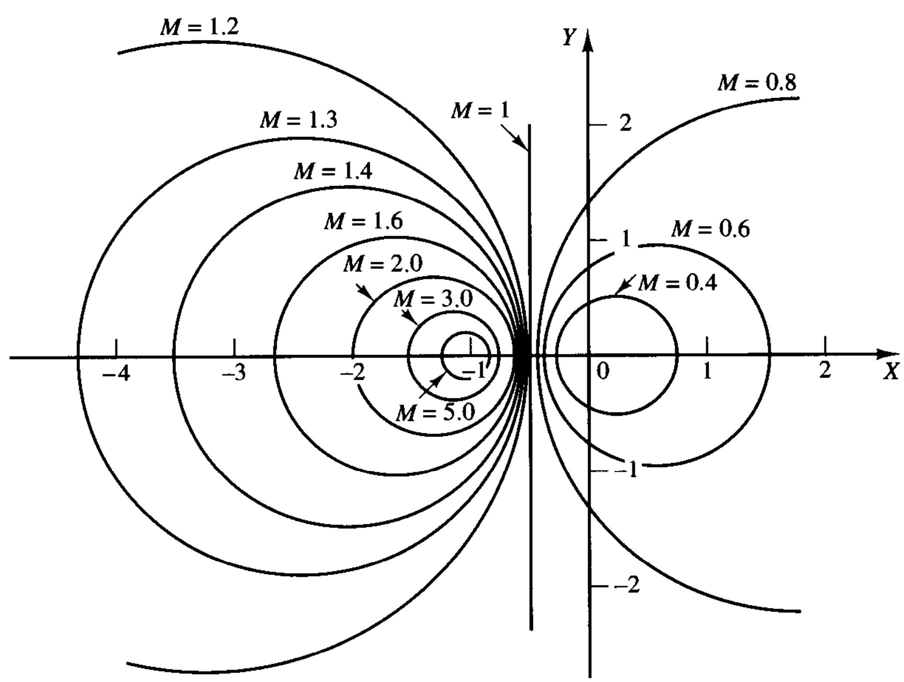

Figure 8–86
A family of constant M circles.

Since

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

we obtain

$$N = \frac{\dfrac{Y}{X} - \dfrac{Y}{1 + X}}{1 + \dfrac{Y}{X}\left(\dfrac{Y}{1 + X}\right)} = \frac{Y}{X^2 + X + Y^2}$$

or

$$X^2 + X + Y^2 - \frac{1}{N}Y = 0$$

The addition of $(\frac{1}{4}) + 1/(2N)^2$ to both sides of this last equation yields

$$\left(X + \frac{1}{2}\right)^2 + \left(Y - \frac{1}{2N}\right)^2 = \frac{1}{4} + \left(\frac{1}{2N}\right)^2 \tag{8–17}$$

This is an equation of a circle with center at $X = -\frac{1}{2}$, $Y = 1/(2N)$ and with radius $\sqrt{(\frac{1}{4}) + 1/(2N)^2}$. For example, if $\alpha = 30°$, then $N = \tan\alpha = 0.577$, and the center and the radius of the circle corresponding to $\alpha = 30°$ are found to be $(-0.5, 0.866)$ and unity, respectively. Since Equation (8–17) is satisfied when $X = Y = 0$ and $X = -1, Y = 0$ regardless of the value of N, each circle passes through the origin and the $-1 + j0$ point. The constant α loci can be drawn easily once the value of N is given. A family of constant N circles is shown in Figure 8–87 with α as a parameter.

It should be noted that the constant N locus for a given value of α is actually not the entire circle but only an arc. In other words, the $\alpha = 30°$ and $\alpha = -150°$ arcs are parts of the same circle. This is so because the tangent of an angle remains the same if $\pm 180°$ (or multiplies thereof) is added to the angle.

The use of the M and N circles enables us to find the entire closed-loop frequency response from the open-loop frequency response $G(j\omega)$ without calculating the magnitude and phase of the closed-loop transfer function at each frequency. The intersections of the $G(j\omega)$ locus and the M circles and N circles gives the values of M and N at frequency points on the $G(j\omega)$ locus.

The N circles are multivalued in the sense that the circle for $\alpha = \alpha_1$ and that for $\alpha = \alpha_1 \pm 180°n$ $(n = 1, 2, \ldots)$ are the same. In using the N circles for the determination of the phase angle of closed-loop systems, we must interpret the proper value of α. To avoid any error, start at zero frequency, which corresponds to $\alpha = 0°$, and proceed to higher frequencies. The phase-angle curve must be continuous.

Graphically, the intersections of the $G(j\omega)$ locus and M circles give the values of M at the frequencies denoted on the $G(j\omega)$ locus. Thus, the constant M circle with the smallest radius that is tangent to the $G(j\omega)$ locus gives the value of the resonant peak magnitude M_r. If it is desired to keep the resonant peak value less than a certain value, then the system should not enclose the critical point ($-1 + j0$ point) and, at the same time, there should be no intersections with the particular M circle and the $G(j\omega)$ locus.

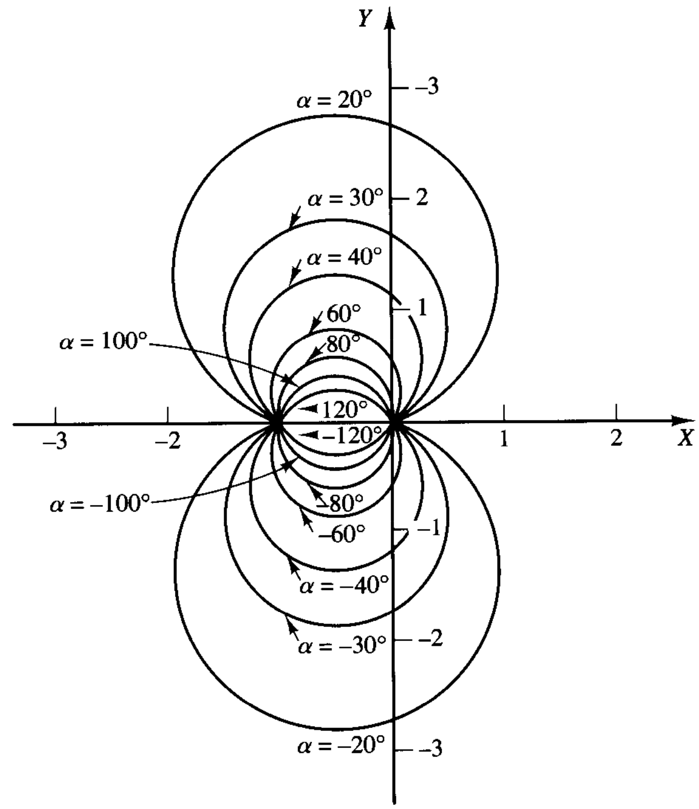

Figure 8–87
A family of constant N circles.

Figure 8–88(a) shows the $G(j\omega)$ locus superimposed on a family of M circles. Figure 8–88(b) shows the $G(j\omega)$ locus superimposed on a family of N circles. From these plots, it is possible to obtain the closed-loop frequency response by inspection. It is seen that the $M = 1.1$ circle intersects the $G(j\omega)$ locus at frequency point $\omega = \omega_1$. This means that at this frequency the magnitude of the closed-loop transfer function is 1.1. In Figure 8–88(a), the $M = 2$ circle is just tangent to the $G(j\omega)$ locus. Thus, there is only one point on the $G(j\omega)$ locus for which $|C(j\omega)/R(j\omega)|$ is equal to 2. Figure 8–88(c) shows the closed-loop frequency-response curve for the system. The upper curve is the M versus frequency ω curve, and the lower curve is the phase angle α versus frequency ω curve.

The resonant peak value is the value of M corresponding to the M circle of smallest radius that is tangent to the $G(j\omega)$ locus. Thus, in the Nyquist diagram, the resonant peak value M_r and the resonant frequency ω_r can be found from the M-circle tangency to the $G(j\omega)$ locus. (In the present example, $M_r = 2$ and $\omega_r = \omega_4$.)

Nichols chart. In dealing with design problems, we find it convenient to construct the M and N loci in the log-magnitude versus phase plane. The chart consisting of the M and N loci in the log-magnitude versus phase diagram is called the Nichols chart. This chart is shown in Figure 8–89, for phase angles between 0° and −240°.

Note that the critical point ($-1 + j0$ point) is mapped to the Nichols chart as the point (0 dB, −180°). The Nichols chart contains curves of constant closed-loop magnitude and phase angle. The designer can graphically determine the phase margin, gain margin, resonant peak magnitude, resonant peak frequency, and bandwidth of the closed-loop system from the plot of the open-loop locus, $G(j\omega)$.

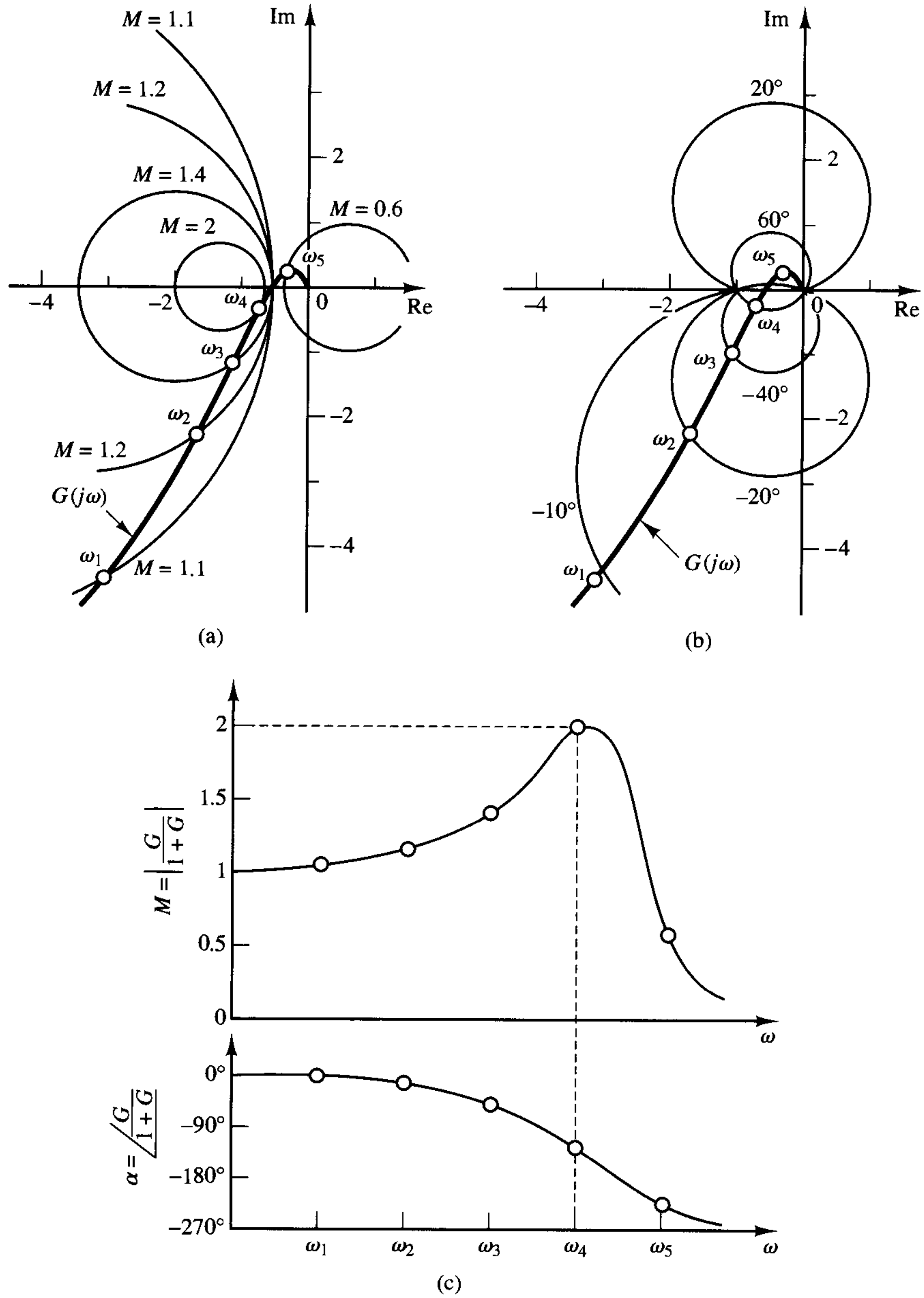

Figure 8–88
(a) $G(j\omega)$ locus superimposed on a family of M circles; (b) $G(j\omega)$ locus superimposed on a family of N circles; (c) closed-loop frequency-response curves.

The Nichols chart is symmetric about the $-180°$ axis. The M and N loci repeat for every 360°, and there is symmetry at every 180° interval. The M loci are centered about the critical point (0 dB, $-180°$). The Nichols chart is useful for determining the frequency response of the closed loop from that of the open loop. If the open-loop frequency-response curve is superimposed on the Nichols chart, the intersections of the open-loop frequency-response curve $G(j\omega)$ and the M and N loci give the values of the magnitude M and phase angle α of the closed-loop frequency response at each frequency point. If

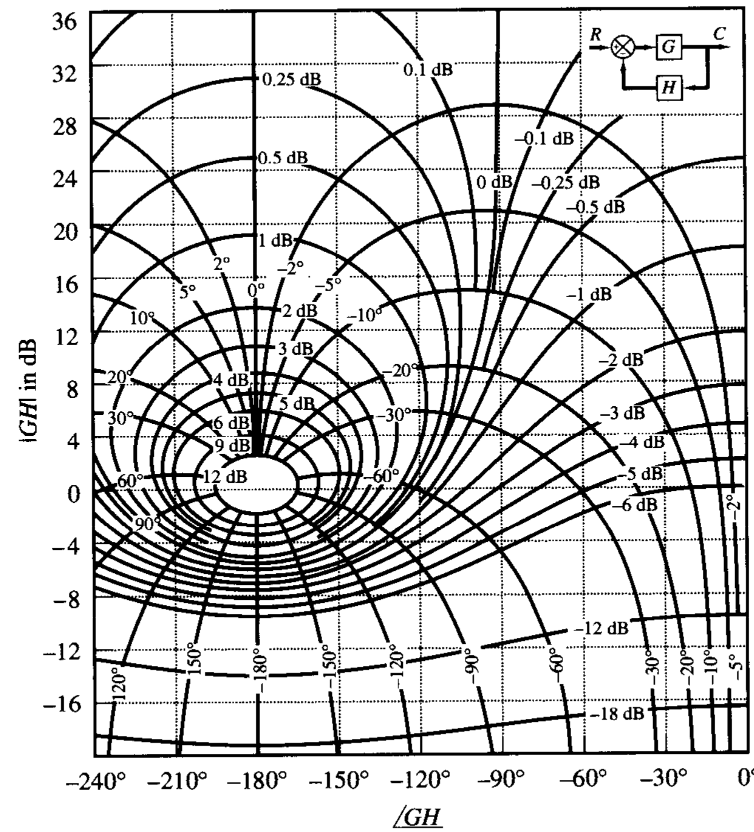

Figure 8–89
Nichols chart.

the $G(j\omega)$ locus does not intersect the $M = M_r$ locus but is tangent to it, then the resonant peak value of M of the closed-loop frequency response is given by M_r. The resonant peak frequency is given by the frequency at the point of tangency.

As an example, consider the unity-feedback system with the following open-loop transfer function:

$$G(j\omega) = \frac{K}{s(s+1)(0.5s+1)}, \qquad K = 1$$

To find the closed-loop frequency response by use of the Nichols chart, the $G(j\omega)$ locus is constructed in the log-magnitude versus phase plane from the Bode diagram. The use of the Bode diagram eliminates the lengthy numerical calculation of $G(j\omega)$. Figure 8–90(a) shows the $G(j\omega)$ locus together with the M and N loci. The closed-loop frequency-response curves may be constructed by reading the magnitudes and phase angles at various frequency points on the $G(j\omega)$ locus from the M and N loci, as shown in Figure 8–90(b). Since the largest magnitude contour touched by the $G(j\omega)$ locus is 5 dB, the resonant peak magnitude M_r is 5 dB. The corresponding resonant peak frequency is 0.8 rad/sec.

Notice that the phase crossover point is the point where the $G(j\omega)$ locus intersects the $-180°$ axis (for the present system, $\omega = 1.4$ rad/sec), and the gain crossover point is the point where the locus intersects the 0-dB axis (for the present system, $\omega = 0.76$

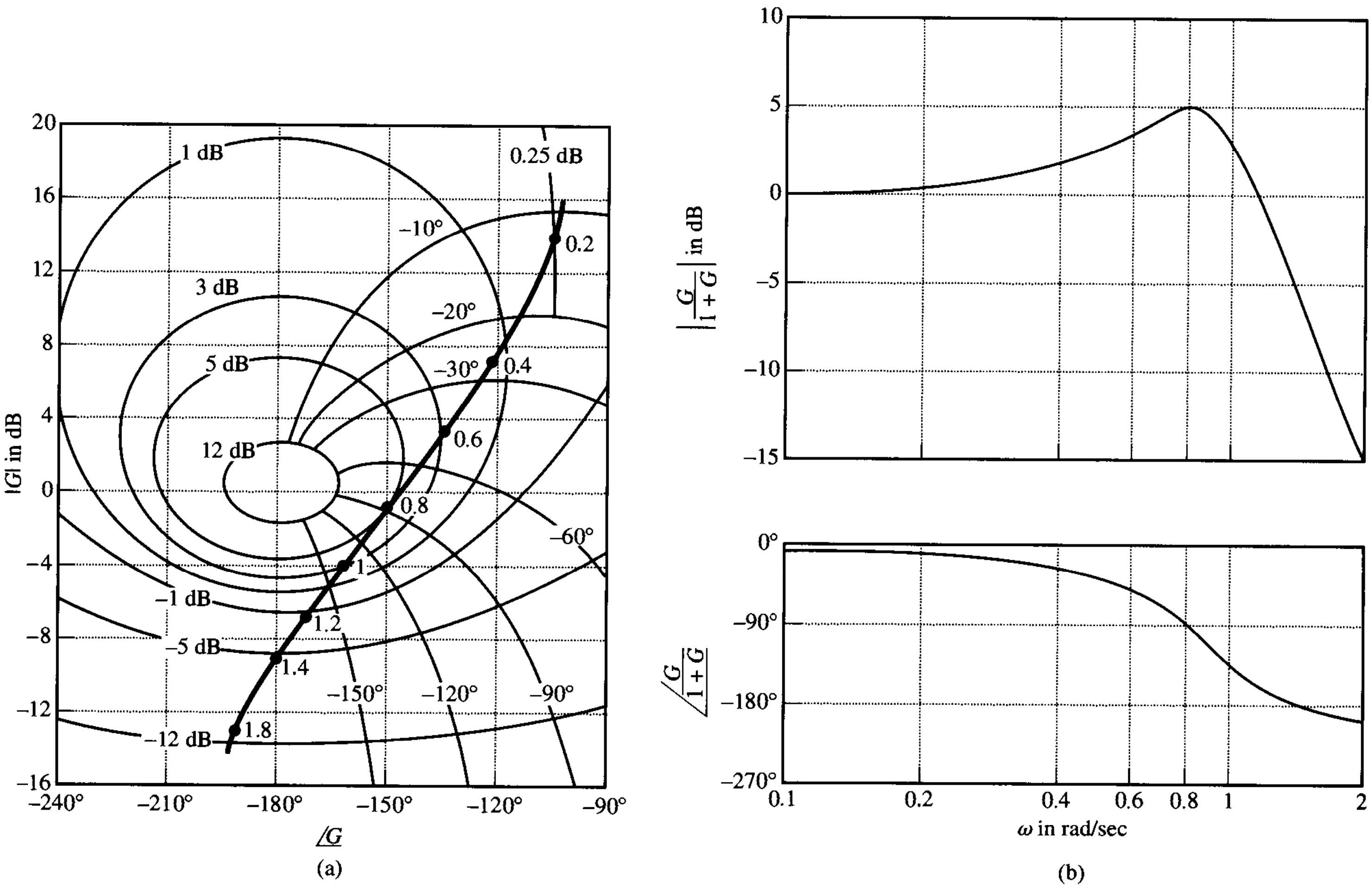

Figure 8–90
(a) Plot of $G(j\omega)$ superimposed on Nichols chart; (b) closed-loop frequency-response curves.

rad/sec). The phase margin is the horizontal distance (measured in degrees) between the gain crossover point and the critical point (0 dB, $-180°$). The gain margin is the distance (in decibels) between the phase crossover point and the critical point.

The bandwidth of the closed-loop system can easily be found from the $G(j\omega)$ locus in the Nichols diagram. The frequency at the intersection of the $G(j\omega)$ locus and the $M = -3$ dB locus gives the bandwidth.

If the open-loop gain K is varied, the shape of the $G(j\omega)$ locus in the log-magnitude versus phase diagram remains the same, but it is shifted up (for increasing K) or down (for decreasing K) along the vertical axis. Therefore, the $G(j\omega)$ locus intersects the M and N loci differently, resulting in a different closed-loop frequency-response curve. For a small value of the gain K, the $G(j\omega)$ locus will not be tangent to any of the M loci, which means that there is no resonance in the closed-loop frequency response.

Closed-loop frequency response for nonunity-feedback systems. In the preceding sections, our discussions were limited to closed-loop systems with unity feedback. The constant M and N loci and the Nichols chart cannot be directly applied to control systems with nonunity feedback, but rather require a slight modification.

If the closed-loop system involves a nonunity-feedback transfer function, then the closed-loop transfer function may be written

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

where $G(s)$ is the feedforward transfer function and $H(s)$ is the feedback transfer function. Then $C(j\omega)/R(j\omega)$ can be written

$$\frac{C(j\omega)}{R(j\omega)} = \frac{1}{H(j\omega)} \frac{G(j\omega)H(j\omega)}{1 + G(j\omega)H(j\omega)}$$

The magnitude and phase angle of

$$\frac{G_1(j\omega)}{1 + G_1(j\omega)}$$

where $G_1(j\omega) = G(j\omega)H(j\omega)$, may be obtained easily by plotting the $G_1(j\omega)$ locus on the Nichols chart and reading the values of M and N at various frequency points. The closed-loop frequency response $C(j\omega)/R(j\omega)$ may then be obtained by multiplying $G_1(j\omega)/[1 + G_1(j\omega)]$ by $1/H(j\omega)$. This multiplication can be made without difficulty if we draw Bode diagrams for $G_1(j\omega)/[1 + G_1(j\omega)]$ and $H(j\omega)$ and then graphically subtract the magnitude of $H(j\omega)$ from that of $G_1(j\omega)/[1 + G_1(j\omega)]$ and also graphically subtract the phase angle of $H(j\omega)$ from that of $G_1(j\omega)/[1 + G_1(j\omega)]$. Then the resulting log-magnitude curve and phase-angle curve give the closed-loop frequency response $C(j\omega)/R(j\omega)$.

To obtain acceptable values of M_r, ω_r, and ω_b for $|C(j\omega)/R(j\omega)|$, a trial-and-error process may be necessary. In each trial, the $G_1(j\omega)$ locus is varied in shape. Then Bode diagrams for $G_1(j\omega)/[1 + G_1(j\omega)]$ and $H(j\omega)$ are drawn, and the closed-loop frequency response $C(j\omega)R(j\omega)$ is obtained. The values of M_r, ω_r, and ω_b are checked until they are acceptable.

Gain adjustments. The concept of M circles will now be applied to the design of control systems. In obtaining suitable performance, the adjustment of gain is usually the first consideration. The adjustment may be based on a desirable value for the resonant peak.

In the following, we shall demonstrate a method for determining the gain K so that the system will have some maximum value M_r, not exceeded over the entire frequency range.

Referring to Figure 8–91, we see that the tangent line drawn from the origin to the desired M_r circle has an angle of ψ, as shown, if M_r is greater than unity. The value of $\sin\psi$ is

$$\sin\psi = \left| \frac{\dfrac{M_r}{M_r^2 - 1}}{\dfrac{M_r^2}{M_r^2 - 1}} \right| = \frac{1}{M_r} \tag{8–18}$$

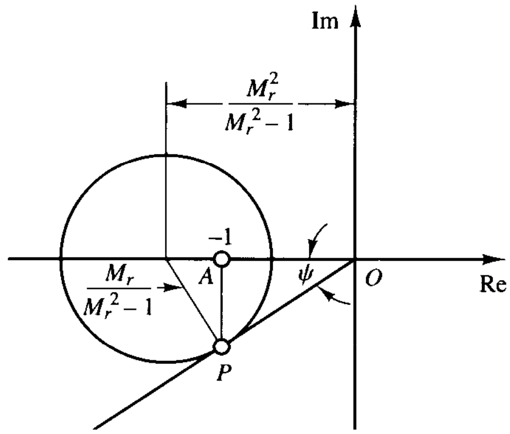

Figure 8–91
M circle.

Let us call the tangency point of the tangent line and the M_r circle as point P. It can easily be proved that the line drawn from point P, perpendicular to the negative real axis, intersects this axis at the $-1 + j0$ point.

Consider the system shown in Figure 8–92. The procedure for determining the gain K so that $G(j\omega) = KG_1(j\omega)$ will have a desired value of M_r (where $M_r > 1$) can be summarized as follows:

1. Draw the polar plot of the normalized open-loop transfer function $G_1(j\omega) = G(j\omega)/K$.
2. Draw from the origin the line that makes an angle of $\psi = \sin^{-1}(1/M_r)$ with the negative real axis.
3. Fit a circle with center on the negative real axis tangent to both the $G_1(j\omega)$ locus and the line PO.
4. Draw a perpendicular line to the negative real axis from point P, the point of tangency of this circle with the line PO. The perpendicular line PA intersects the negative real axis at point A.
5. For the circle just drawn to correspond to the desired M_r circle, point A should be the $-1 + j0$ point.
6. The desired value of the gain K is that value which changes the scale so that point A becomes the $-1 + j0$ point. Thus, $K = 1/\overline{OA}$.

Note that the resonant frequency ω_r is the frequency of the point at which the circle is tangent to the $G_1(j\omega)$ locus. The present procedure may not yield a satisfactory value for ω_r. If this is the case, the system must be compensated in order to increase the value

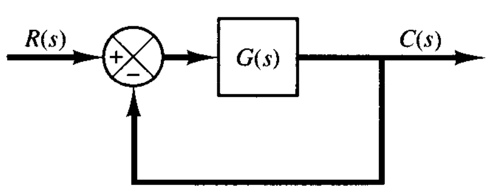

Figure 8–92
Control system.

of ω_r without changing the value of M_r. (For the compensation of control systems by frequency-response methods, see Chapter 9.)

Note also that if the system has nonunity feedback then the method requires some cut-and-try steps.

EXAMPLE 8–22 Consider the unity-feedback control system whose open-loop transfer function is

$$G(j\omega) = \frac{K}{j\omega(1 + j\omega)}$$

Determine the value of the gain K so that $M_r = 1.4$.

The first step in the determination of the gain K is to sketch the polar plot of

$$\frac{G(j\omega)}{K} = \frac{1}{j\omega(1 + j\omega)}$$

as shown in Figure 8–93. The value of ψ corresponding to $M_r = 1.4$ is obtained from

$$\psi = \sin^{-1}\frac{1}{M_r} = \sin^{-1}\frac{1}{1.4} = 45.6°$$

The next step is to draw the line OP that makes an angle $\psi = 45.6°$ with the negative real axis. Then draw the circle that is tangent to both the $G(j\omega)/K$ locus and the line OP. Define the point where the circle is tangent to the 45.6° line as point P. The perpendicular line drawn from point P intersects the negative real axis at $(-0.63, 0)$. Then the gain K of the system is determined as follows:

$$K = \frac{1}{0.63} = 1.59$$

It should be noted that such determination of the gain can also be done easily on the log-magnitude versus phase plot. In what follows, we shall demonstrate how the log-magnitude versus phase diagram can be used to determine the gain K so that the system will have a desired value of M_r.

Figure 8–94 shows the $M_r = 1.4$ locus and the $G(j\omega)/K$ locus. Changing the gain has no effect on the phase angle but merely moves the curve vertically up for $K > 1$ and down for $K < 1$.

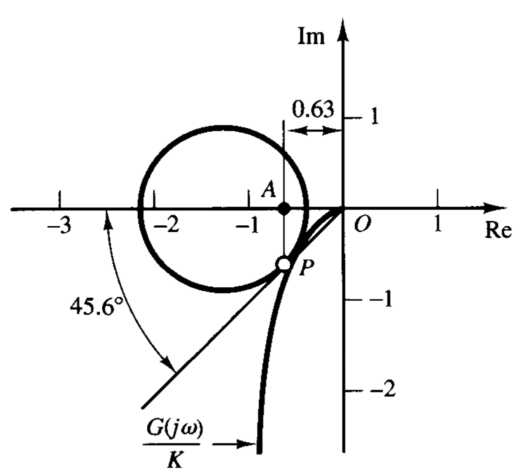

Figure 8–93
Determination of the gain K using an M circle.

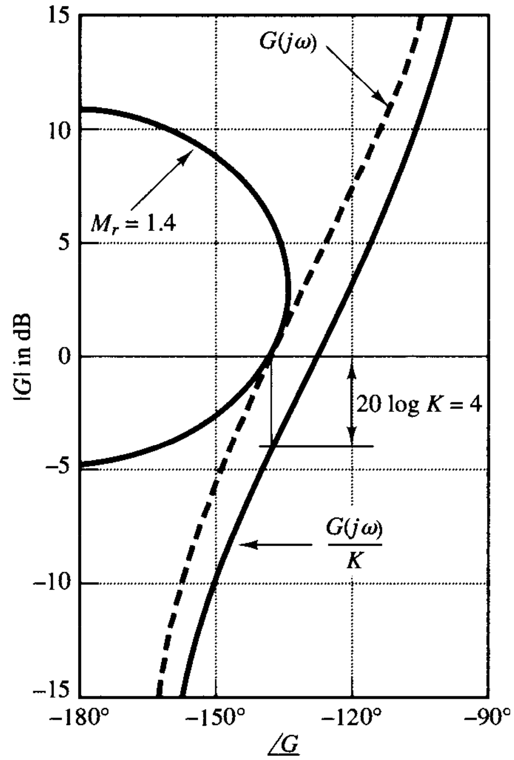

Figure 8–94
Determination of the gain K using the Nichols chart.

In Figure 8–94, the $G(j\omega)/K$ locus must be raised by 4 dB in order that it be tangent to the desired M_r locus and that the entire $G(j\omega)/K$ locus be outside the $M_r = 1.4$ locus. The amount of vertical shift of the $G(j\omega)/K$ locus determines the gain necessary to yield the desired value of M_r. Thus, by solving

$$20 \log K = 4$$

we obtain

$$K = 1.59$$

Thus, we have the same result as obtained earlier.

8–11 EXPERIMENTAL DETERMINATION OF TRANSFER FUNCTIONS

The first step in the analysis and design of a control system is to derive a mathematical model of the plant under consideration. Obtaining a model analytically may be quite difficult. We may have to obtain it by means of experimental analysis. The importance of the frequency-response methods is that the transfer function of the plant, or any other component of a system, may be determined by simple frequency-response measurements.

If the amplitude ratio and phase shift have been measured at a sufficient number of frequencies within the frequency range of interest, they may be plotted on the Bode diagram. Then the transfer function can be determined by asymptotic approximations. We build up asymptotic log-magnitude curves consisting of several segments. With some trial-and-error juggling of the corner frequencies, it is usually possible to find a very close fit to the curve. (Note that if the frequency is plotted in cycles per second rather than radians per second the corner frequencies must be converted to radians per second before computing the time constants.)

Sinusoidal-signal generators. In performing a frequency-response test, suitable sinusoidal-signal generators must be available. The signal may have to be in mechanical, electrical, or pneumatic form. The frequency ranges needed for the test are approximately 0.001 to 10 Hz for large-time-constant systems and 0.1 to 1000 Hz for small-time-constant systems. The sinusoidal signal must be reasonably free from harmonics or distortion.

For very low frequency ranges (below 0.01 Hz) a mechanical signal generator (together with a suitable pneumatic or electrical transducer if necessary) may be used. For the frequency range from 0.01 to 1000 Hz, a suitable electrical-signal generator (together with a suitable transducer if necessary) may be used.

Determination of minimum-phase transfer functions from Bode diagrams. As stated previously, whether a system is minimum phase can be determined from the frequency-response curves by examining the high-frequency characteristics.

To determine the transfer function, we first draw asymptotes to the experimentally obtained log-magnitude curve. The asymptotes must have slopes of multiples of ± 20 dB/decade. If the slope of the experimentally obtained log-magnitude curve changes from -20 to -40 dB/decade at $\omega = \omega_1$, it is clear that a factor $1/[1 + j(\omega/\omega_1)]$ exists in the transfer function. If the slope changes by -40 dB/decade at $\omega = \omega_2$, there must be a quadratic factor of the form

$$\frac{1}{1 + 2\zeta\left(j\dfrac{\omega}{\omega_2}\right) + \left(j\dfrac{\omega}{\omega_2}\right)^2}$$

in the transfer function. The undamped natural frequency of this quadratic factor is equal to the corner frequency ω_2. The damping ratio ζ can be determined from the experimentally obtained log-magnitude curve by measuring the amount of resonant peak near the corner frequency ω_2 and comparing this with the curves shown in Figure 8–8.

Once the factors of the transfer function $G(j\omega)$ have been determined, the gain can be determined from the low-frequency portion of the log-magnitude curve. Since such terms as $1 + j(\omega/\omega_1)$ and $1 + 2\,\zeta(j\omega/\omega_2 + (j\omega/\omega_2)^2$ become unity as ω approaches zero, at very low frequencies, the sinusoidal transfer function $G(j\omega)$ can be written

$$\lim_{\omega \to 0} G(j\omega) = \frac{K}{(j\omega)^\lambda}$$

In many practical systems, λ equals 0, 1, or 2.

1. For $\lambda = 0$, or type 0 systems,

$$G(j\omega) = K, \qquad \text{for } \omega \ll 1$$

or

$$20 \log |G(j\omega)| = 20 \log K, \qquad \text{for } \omega \ll 1$$

The low-frequency asymptote is a horizontal line at $20 \log K$ dB. The value of K can thus be found from this horizontal asymptote.

2. For $\lambda = 1$, or type 1 systems,

$$G(j\omega) = \frac{K}{j\omega}, \qquad \text{for } \omega \ll 1$$

or

$$20 \log |G(j\omega)| = 20 \log K - 20 \log \omega, \qquad \text{for } \omega \ll 1$$

which indicates that the low-frequency asymptote has the slope -20 dB/decade. The frequency at which the low-frequency asymptote (or its extension) intersects the 0-dB line is numerically equal to K.

3. For $\lambda = 2$, or type 2 systems,

$$G(j\omega) = \frac{K}{(j\omega)^2}, \qquad \text{for } \omega \ll 1$$

or

$$20 \log |G(j\omega)| = 20 \log K - 40 \log \omega, \qquad \text{for } \omega \ll 1$$

The slope of the low-frequency asymptote is -40 dB/decade. The frequency at which this asymptote (or its extension) intersects the 0-dB line is numerically equal to $\sqrt{K}$.

Examples of log-magnitude curves for type 0, type 1, and type 2 systems are shown in Figure 8–95, together with the frequency to which the gain K is related.

The experimentally-obtained phase-angle curve provides a means of checking the transfer function obtained from the log-magnitude curve. For a minimum-phase system, the experimental phase-angle curve should agree reasonably well with the theoretical phase-angle curve obtained from the transfer function just determined. These two phase-angle curves should agree exactly in both the very low and very high frequency ranges. If the experimentally obtained phase angle at very high frequencies (compared with the corner frequencies) is not equal to $-90°(q - p)$, where p and q are the degrees of the numerator and denominator polynomials of the transfer function, respectively, then the transfer function must be a nonminimum-phase transfer function.

Nonminimum-phase transfer functions. If, at the high-frequency end, the computed phase lag is 180° less than the experimentally obtained phase lag, then one of the zeros of the transfer function should have been in the right-half s plane instead of the left-half s plane.

If the computed phase lag differed from the experimentally obtained phase lag by a constant rate of change of phase, then transport lag, or dead time, is present. If we assume the transfer function to be of the form

$$G(s)e^{-Ts}$$

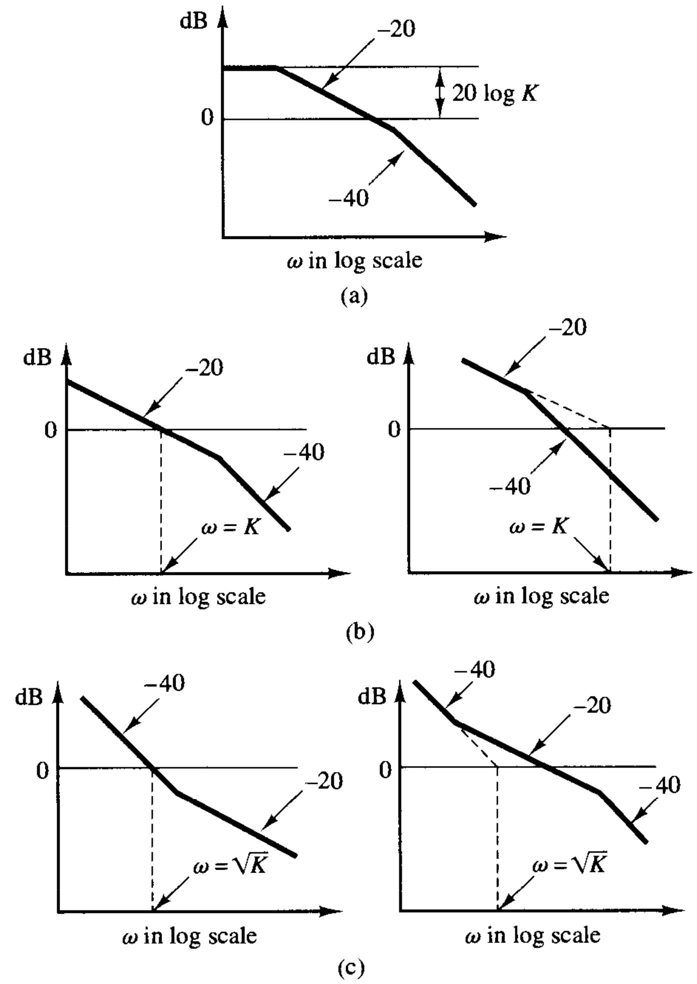

Figure 8–95
(a) Log-magnitude curve of a type 0 system; (b) log-magnitude curves of type 1 systems; (c) log-magnitude curves of type 2 systems. (The slopes shown are in dB/decade.)

where $G(s)$ is a ratio of two polynomials in s, then

$$\begin{aligned}\lim_{\omega\to\infty}\frac{d}{d\omega}\underline{/G(j\omega)e^{-j\omega T}} &= \lim_{\omega\to\infty}\frac{d}{d\omega}\left[\underline{/G(j\omega)} + \underline{/e^{-j\omega T}}\right]\\ &= \lim_{\omega\to\infty}\frac{d}{d\omega}\left[\underline{/G(j\omega)} - \omega T\right]\\ &= 0 - T = -T\end{aligned}$$

Thus, from this last equation, we can evaluate the magnitude of the transport lag T.

A few remarks on the experimental determination of transfer functions

1. It is usually easier to make accurate amplitude measurements than accurate phase-shift measurements. Phase-shift measurements may involve errors that may be caused by instrumentation or by misinterpretation of the experimental records.

2. The frequency response of measuring equipment used to measure the system output must have a nearly flat magnitude versus frequency curves. In addition, the phase angle must be nearly proportional to the frequency.

3. Physical systems may have several kinds of nonlinearities. Therefore, it is necessary to consider carefully the amplitude of input sinusoidal signals. If the amplitude of the input signal is too large, the system will saturate, and the frequency-response test will yield inaccurate results. On the other hand, a small signal will cause errors due to dead zone. Hence, a careful choice of the amplitude of the input sinusoidal signal must be made. It is necessary to sample the waveform of the system output to make sure that the waveform is sinusoidal and that the system is operating in the linear region during the test period. (The waveform of the system output is not sinusoidal when the system is operating in its nonlinear region.)

4. If the system under consideration is operating continuously for days and weeks, then normal operation need not be stopped for frequency-response tests. The sinusoidal test signal may be superimposed on the normal inputs. Then, for linear systems, the output due to the test signal is superimposed on the normal output. For the determination of the transfer function while the system is in normal operation, stochastic signals (white noise signals) also are often used. By use of correlation functions, the transfer function of the system can be determined without interrupting normal operation.

EXAMPLE 8–23 Determine the transfer function of the system whose experimental frequency-response curves are as shown in Figure 8–96.

The first step in determining the transfer function is to approximate the log-magnitude curve by asymptotes with slopes ±20 dB/decade and multiples thereof, as shown in Figure 8–96. We then estimate the corner frequencies. For the system shown in Figure 8–96, the following form of the transfer function is estimated.

$$G(j\omega) = \frac{K(1 + 0.5j\omega)}{j\omega(1 + j\omega)\left[1 + 2\zeta\left(j\frac{\omega}{8}\right) + \left(j\frac{\omega}{8}\right)^2\right]}$$

The value of the damping ratio ζ is estimated by examining the peak resonance near $\omega = 6$ rad/sec. Referring to Figure 8–8, ζ is determined to be 0.5. The gain K is numerically equal to the frequency at the intersection of the extension of the low-frequency asymptote and the 0-dB line. The value of K is thus found to be 10. Therefore, $G(j\omega)$ is tentatively determined as

$$G(j\omega) = \frac{10(1 + 0.5j\omega)}{j\omega(1 + j\omega)\left[1 + \left(j\frac{\omega}{8}\right) + \left(j\frac{\omega}{8}\right)^2\right]}$$

or

$$G(s) = \frac{320(s + 2)}{s(s + 1)(s^2 + 8s + 64)}$$

This transfer function is tentative because we have not examined the phase-angle curve yet.

Once the corner frequencies are noted on the log-magnitude curve, the corresponding phase-angle curve for each component factor of the transfer function can easily be drawn.

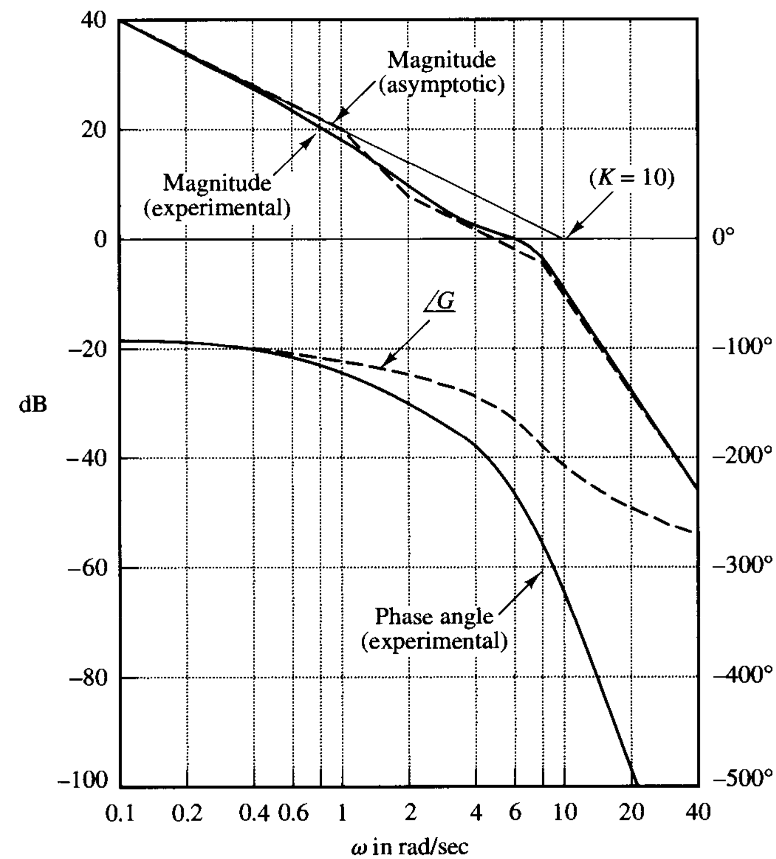

Figure 8–96
Bode diagram of a system. (Solid curves are experimentally obtained curves.)

The sum of these component phase-angle curves is that of the assumed transfer function. The phase-angle curve for $G(j\omega)$ is denoted by $\underline{/G}$ in Figure 8–96. From Figure 8–96 we clearly notice a discrepancy between the computed phase-angle curve and the experimentally-obtained phase-angle curve. The difference between the two curves at very high frequencies appears to be a constant rate of change. Thus, the discrepancy in the phase-angle curves must be caused by transport lag.

Hence, we assume the complete transfer function to be $G(s)e^{-Ts}$. Since the discrepancy between the computed and experimental phase angles is -0.2ω rad for very high frequencies, we can determine the value of T as follows.

$$\lim_{\omega \to \infty} \frac{d}{d\omega} \underline{/G(j\omega)e^{-j\omega T}} = -T = -0.2$$

or

$$T = 0.2 \text{ sec}$$

The presence of transport lag can thus be determined, and the complete transfer function determined from the experimental curves is

$$G(s)e^{-Ts} = \frac{320(s+2)e^{-0.2s}}{s(s+1)(s^2+8s+64)}$$

A-8-1. Consider a system whose closed-loop transfer function is

$$\frac{C(s)}{R(s)} = \frac{10(s+1)}{(s+2)(s+5)}$$

(This is the same system considered in Problem A-6-9.) Clearly, the closed-loop poles are located at $s = -2$ and $s = -5$, and the system is not oscillatory. (The unit-step response, however, exhibits overshoot due to the presence of a zero at $s = -1$. See Figure 6–51.)

Show that the closed-loop frequency response of this system will exhibit a resonant peak, although the damping ratio of the closed-loop poles is greater then unity.

Solution. Figure 8–97 shows the Bode diagram for the system. The resonant peak value is approximately 3.5 dB. (Note that, in the absence of a zero, the second-order system with $\zeta > 0.7$ will not exhibit a resonant peak; however, the presence of a closed-loop zero will cause such a peak.)

A-8-2. Plot a Bode diagram for the following open-loop transfer function $G(s)$:

$$G(s) = \frac{20(s^2 + s + 0.5)}{s(s+1)(s+10)}$$

Solution. By substituting $s = j\omega$ into $G(s)$, we have

$$G(j\omega) = \frac{20[(j\omega)^2 + (j\omega) + 0.5]}{j\omega(j\omega+1)(j\omega+10)}$$

Notice that ω_n and ζ of the quadratic term in the numerator are

$$\omega_n = \sqrt{0.5} \qquad \text{and} \qquad \zeta = 0.707$$

This quadratic term can be written as

$$\omega_n^2\left[\left(j\frac{\omega}{\omega_n}\right)^2 + 2\zeta\left(j\frac{\omega}{\omega_n}\right) + 1\right] = (\sqrt{0.5})^2\left[\left(j\frac{\omega}{\sqrt{0.5}}\right)^2 + 2 \times 0.707\left(j\frac{\omega}{\sqrt{0.5}}\right) + 1\right]$$

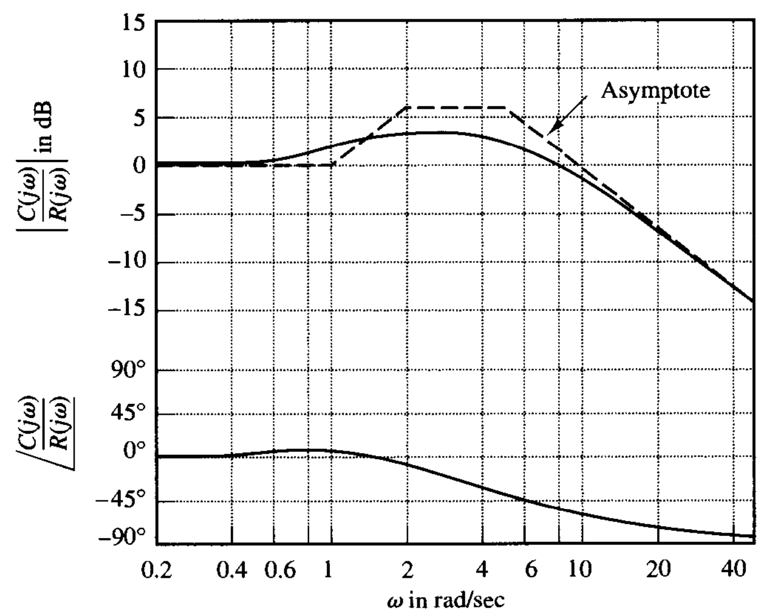

Figure 8–97 Bode diagram for $10(1 + j\omega)/[(2 + j\omega)(5 + j\omega)]$

Note that the corner frequency is at $\omega = \sqrt{0.5} = 0.707$ rad/sec. Now $G(j\omega)$ can be written as

$$G(j\omega) = \frac{\left(j\frac{\omega}{\sqrt{0.5}}\right)^2 + 1.414\left(j\frac{\omega}{\sqrt{0.5}}\right) + 1}{j\omega(j\omega + 1)(0.1j\omega + 1)}$$

The Bode diagram for $G(j\omega)$ is shown in Figure 8–98.

A-8-3. Draw the Bode diagram of the following nonminimum-phase system:

$$\frac{C(s)}{R(s)} = 1 - Ts$$

Obtain the unit-ramp response of the system and plot $c(t)$ versus t.

Solution. The Bode diagram of the system is shown in Figure 8–99. For a unit-ramp input, $R(s) = 1/s^2$, we have

$$C(s) = \frac{1 - Ts}{s^2} = \frac{1}{s^2} - \frac{T}{s}$$

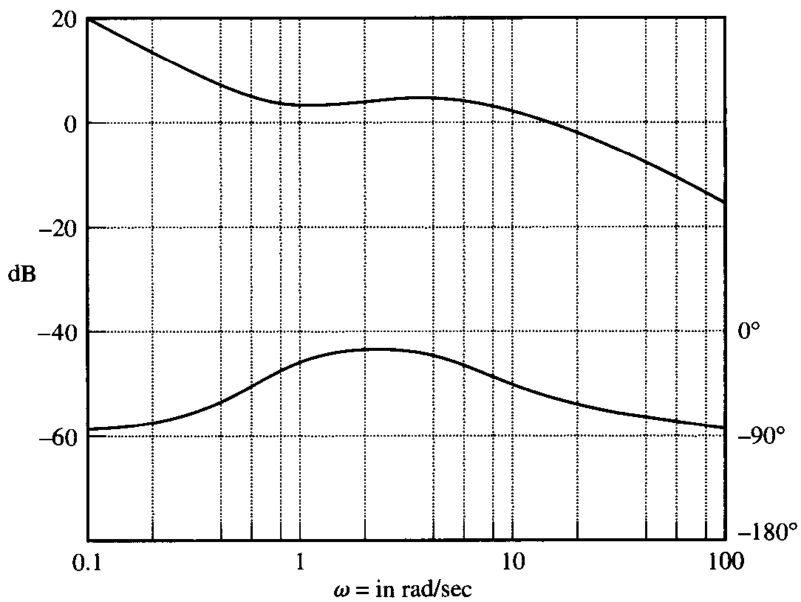

Figure 8–98
Bode diagram for $G(j\omega)$ of Problem A-8-2.

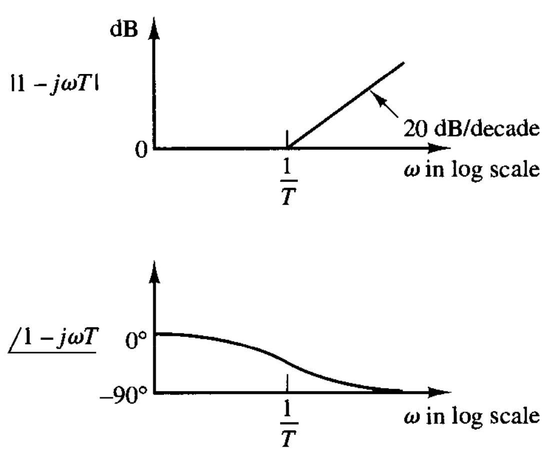

Figure 8–99
Bode diagram of $1 - j\omega T$.

The inverse Laplace transform of $C(s)$ gives

$$c(t) = t - T, \qquad \text{for } t \geq 0$$

Figure 8–100 shows the response curve $c(t)$ versus t. (Note the faulty behavior at the start of the response.) A characteristic property of such a nonminimum-phase system is that the transient response starts out in the opposite direction to the input but eventually comes back in the same direction.

A-8-4. Consider the system defined by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -25 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Obtain the sinusoidal transfer functions $Y_1(j\omega)/U_1(j\omega)$, $Y_2(j\omega)/U_1(j\omega)$, $Y_1(j\omega)/U_2(j\omega)$, and $Y_2(j\omega)/U_2(j\omega)$. In deriving $Y_1(j\omega)/U_1(j\omega)$ and $Y_2(j\omega)/U_1(j\omega)$, we assume that $U_2(j\omega) = 0$. Similarly, in obtaining $Y_1(j\omega)/U_2(j\omega)$and $Y_2(j\omega)/U_2(j\omega)$, we assume that $U_1(j\omega) = 0$. Obtain also the Bode diagrams of these four transfer functions with MATLAB.

Solution. The transfer matrix expression for the system defined by

$$\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu}$$

$$\dot{\mathbf{y}} = \mathbf{Cx} + \mathbf{Du}$$

is given by

$$\mathbf{Y}(s) = \mathbf{G}(s)\mathbf{U}(s)$$

where $\mathbf{G}(s)$ is the transfer matrix and is given by

$$\mathbf{G}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$$

For the system considered here, the transfer matrix becomes

$$\begin{aligned} \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} s & -1 \\ 25 & s+4 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \\ &= \frac{1}{s^2 + 4s + 25} \begin{bmatrix} s+4 & 1 \\ -25 & s \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \dfrac{s+4}{s^2+4s+25} & \dfrac{s+5}{s^2+4s+25} \\ \dfrac{-25}{s^2+4s+25} & \dfrac{s-25}{s^2+4s+25} \end{bmatrix} \end{aligned}$$

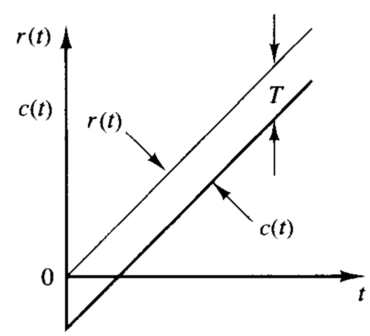

Figure 8–100
Unit-ramp response of the system considered in Problem A-8-3.

Hence

$$\begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} \dfrac{s+4}{s^2+4s+25} & \dfrac{s+5}{s^2+4s+25} \\ \dfrac{-25}{s^2+4s+25} & \dfrac{s-25}{s^2+4s+25} \end{bmatrix} \begin{bmatrix} U_1(s) \\ U_2(s) \end{bmatrix}$$

Assuming that $U_2(j\omega) = 0$, we find $Y_1(j\omega)/U_1(j\omega)$ and $Y_2(j\omega)/U_1(j\omega)$ as follows:

$$\frac{Y_1(j\omega)}{U_1(j\omega)} = \frac{j\omega + 4}{(j\omega)^2 + 4j\omega + 25}$$

$$\frac{Y_2(j\omega)}{U_1(j\omega)} = \frac{-25}{(j\omega)^2 + 4j\omega + 25}$$

Similarly, assuming that $U_1(j\omega) = 0$, we find $Y_1(j\omega)/U_2(j\omega)$ and $Y_2(j\omega)/U_2(j\omega)$ as follows:

$$\frac{Y_1(j\omega)}{U_2(j\omega)} = \frac{j\omega + 5}{(j\omega)^2 + 4j\omega + 25}$$

$$\frac{Y_2(j\omega)}{U_2(j\omega)} = \frac{j\omega - 25}{(j\omega)^2 + 4j\omega + 25}$$

Notice that $Y_2(j\omega)/U_2(j\omega)$ is a nonminimum-phase transfer function.

To plot Bode diagrams for $Y_1(j\omega)/U_1(j\omega)$, $Y_2(j\omega)/U_1(j\omega)$, $Y_1(j\omega)/U_2(j\omega)$, and $Y_2(j\omega)/U_2(j\omega)$ with MATLAB, we may use the command

```
bode(A,B,C,D)
```

Then MATLAB produces Bode diagrams when u_1 is the input and u_2 is zero and when u_2 is the input and u_1 is zero. See MATLAB Program 8–14 and the resulting Bode diagrams shown in Figure 8–101. [Note that MATLAB produces two sets of figures (called Figure 1 and Figure 2) on the screen. Figure 8–101 consists of these two sets of Bode diagrams.]

MATLAB Program 8–14

```
A = [0  1;-25  -4];
B = [1  1;0  1];
C = [1  0;0  1];
D = [0  0;0  0];
bode(A,B,C,D)
```

A-8-5. Referring to Problem A-8-4, consider an alternative way to plot Bode diagrams of the system. One way to plot the Bode diagrams is to use the command

```
bode(A,B,C,D,1)
```

to obtain Bode diagrams for $Y_1(j\omega)/U_1(j\omega)$ and $Y_2(j\omega)/U_1(j\omega)$. To obtain Bode diagrams for $Y_1(j\omega)/U_2(j\omega)$ and $Y_2(j\omega)/U_2(j\omega)$, use the command

```
bode(A,B,C,D,2)
```

Write a MATLAB program to obtain Bode diagrams using these bode commands.

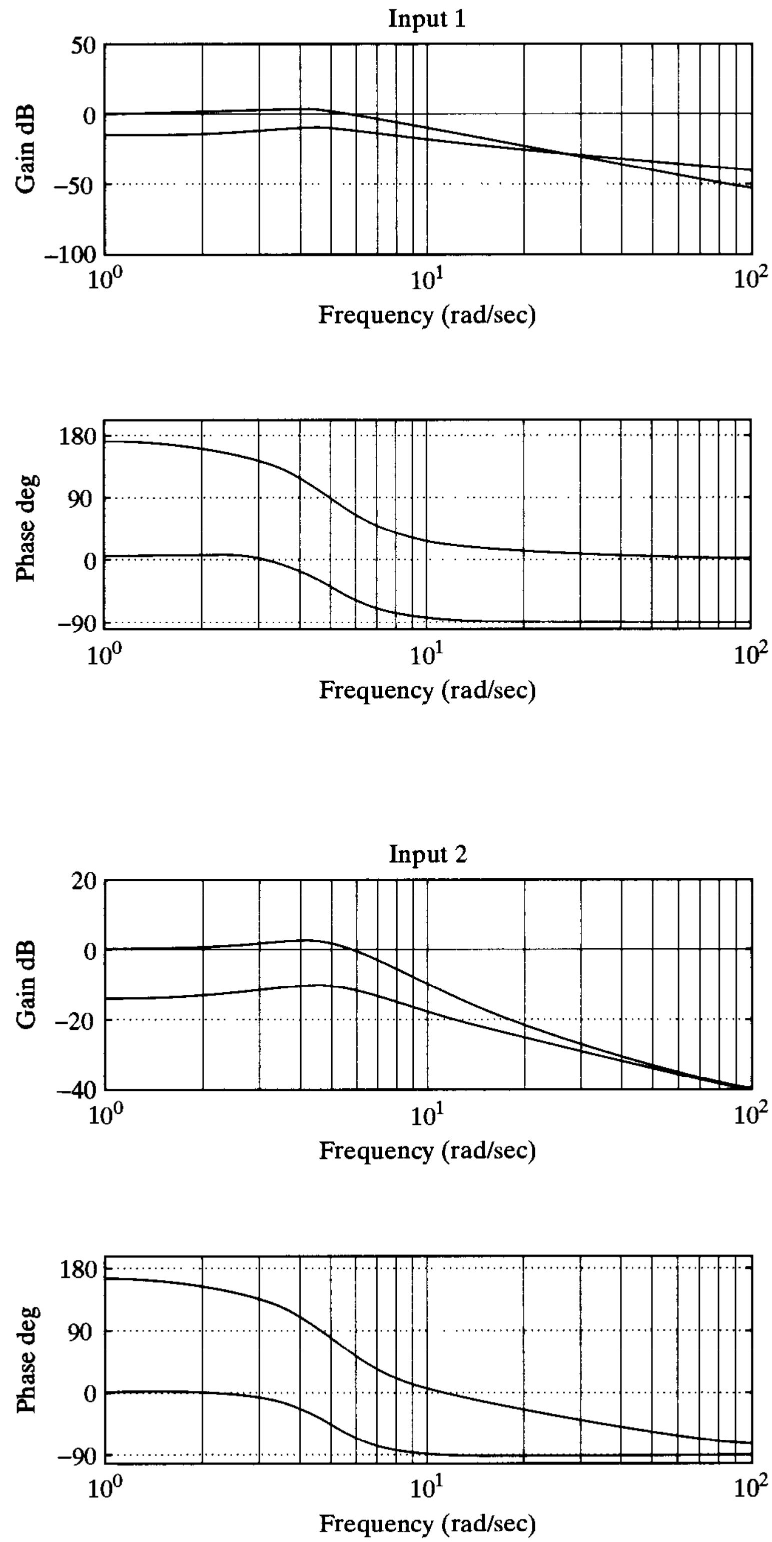

Figure 8–101
Bode diagrams of the system considered in Problem A-8-4.

```
MATLAB Program 8-15

A = [0   1;-25   -4];
B = [1   1;0   1];
C = [1   0;0   1];
D = [0   0;0   0];
bode(A,B,C,D,1)
subplot(2,1,1), title('Bode Diagrams; Input = u1 (u2 = 0)')
bode(A,B,C,D,2)
subplot(2,1,1), title ('Bode Diagrams; Input = u2 (u1 = 0)')
```

Solution. The program for this problem is given as MATLAB Program 8–15. The Bode diagrams produced by this program are shown in Figure 8–102. In these diagrams it may not be easy to identify which curves are for $Y_1(j\omega)$ or $Y_2(j\omega)$. Ordinarily the text command may be used to identify curves. However, the text command does not apply to the present bode commands. To use the text command, we may use the following command:

```
[mag,phase,w] = bode(A,B,C,D,iu,w)
```

See Problem A-8-6 for the details.

A-8-6. Referring to Problems A-8-4 and A-8-5, consider plotting Bode diagrams of the same system as discussed in these problems. Use the text command to distinguish curves in the diagrams. Write a possible MATLAB program for plotting Bode diagrams using the following command:

```
[mag,phase,w] = bode(A,B,C,D,iu,w)
```

Solution. In using the command specified, note that the matrices mag and phase contain the magnitudes of $Y_1(j\omega)$ and $Y_2(j\omega)$ and phase angles for $Y_1(j\omega)$ and $Y_2(j\omega)$ evaluated at each frequency point considered. To get the magnitude of $Y_1(j\omega)$, use the following command:

```
Y1 = mag*[1;0]
```

To convert the magnitude to decibels, use the statement

```
magdB = 20*log10(mag)
```

Hence, to convert Y1 to decibels, enter the statement

```
Y1dB = 20*log10(Y1)
```

Similarly, to plot the magnitude of Y2 in decibels, use the following command:

```
Y2 = mag*[0;1]
Y2dB = 20*log10(Y2)
```

Then enter the command

```
semilogx(w,Y1dB,'o',w,Y1dB,'-',w,Y2dB,'x',w,Y2dB,'-')
```

The text command can now be used to write text in the figure. See MATLAB Program 8–16.

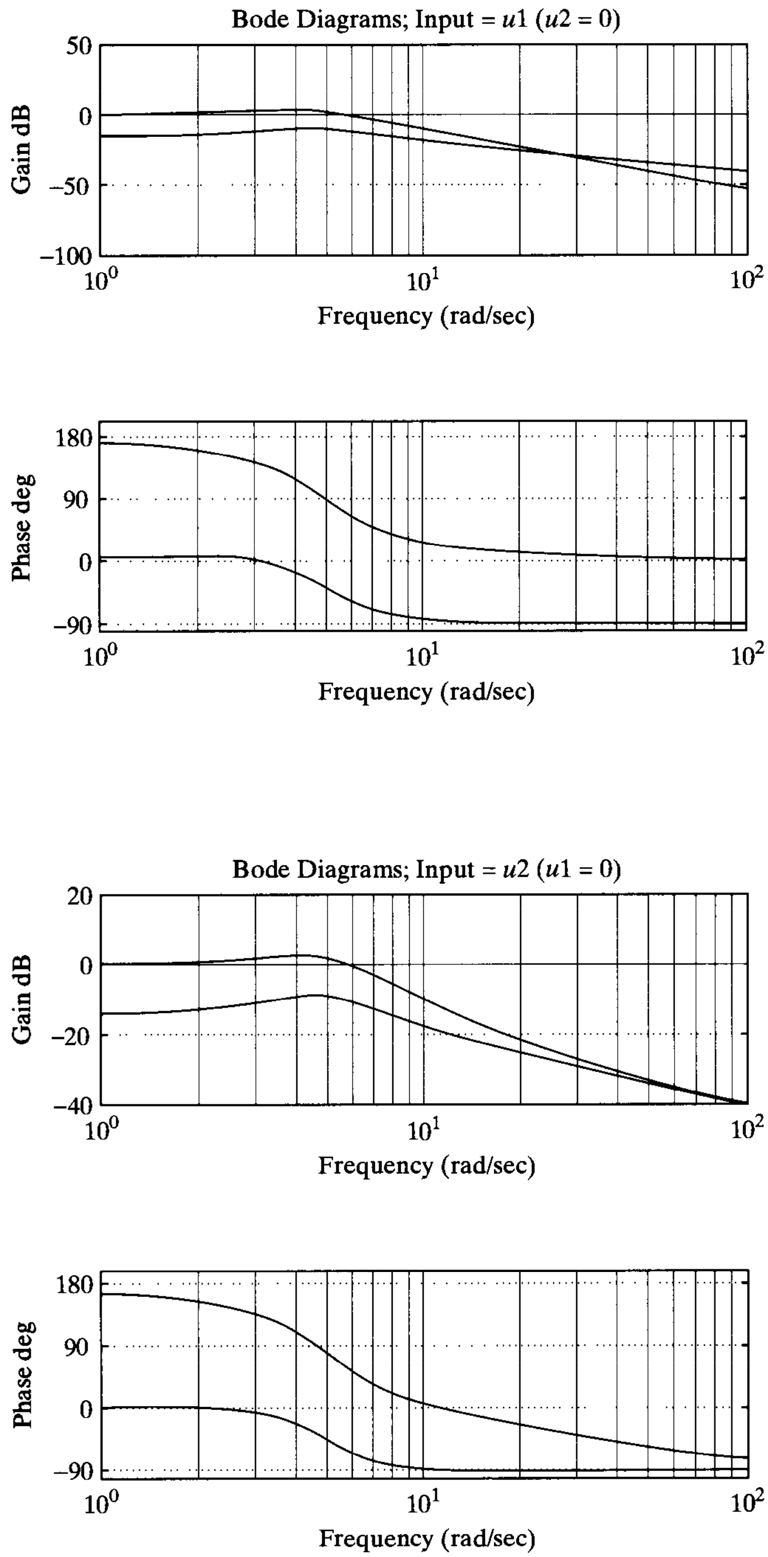

Figure 8–102
Bode diagrams.

MATLAB Program 8–16

```
% ***** We shall first obtain the frequency response when
% u2 = 0; that is, we obtain Y1(jw)/U1(jw) and Y2(jw)/U1(jw) *****

A = [0  1;-25  -4];
B = [1  1;0  1];
C = [1  0;0  1];
D = [0  0;0  0];
w = logspace(-1,3,100);
[mag1, phase1, w] = bode(A,B,C,D,1,w);
Y1 = mag1*[1;0]; Y1dB = 20*log10(Y1);
Y2 = mag1*[0;1]; Y2dB = 20*log10(Y2);
semilogx(w,Y1dB,'o',w,Y1dB,'-',w,Y2dB,'x',w,Y2dB,'-')
grid
subplot(2,1,1);
title('Bode Diagrams: Input = u1 (u2 = 0)')
xlabel('Frequency (rad/sec)')
ylabel('Gain dB')
text(1.2,-10,'Y1'); text(1.2,6,'Y2')

Y1p = phase1*[1;0];
Y2p = phase1*[0;1];
semilogx(w,Y1p,'o',w,Y1p,'-',w,Y2p,'x',w,Y2p,'-')
grid
xlabel('Frequency (rad/sec)')
ylabel('Phase deg')
text(1.2,25,'Y1');text(1.2,188,'Y2')

% ***** In the following, we shall obtain the frequency response
% when u1 = 0; that is, we obtain Y1(jw)/U2(jw) and Y2(jw)/U2(jw) *****

[mag2,phase2,w] = bode(A,B,C,D,2,w);
YY1 = mag2*[1;0]; YY1dB = 20*log10(YY1);
YY2 = mag2*[0;1]; YY2dB = 20*log10(YY2);
semilogx(w,YY1dB,'o',w,YY1dB,'-',w,YY2dB,'x',w,YY2dB,'-')
grid
subplot(2,1,1);
title('Bode Diagrams: Input = u2 (u1 = 0)')
xlabel('Frequency (rad/sec)')
ylabel('Gain dB')
text(1.2, -17,'Y1'); text(1.2,5,'Y2')

YY1p = phase2*[1;0];
YY2p = phase2*[0;1];
semilogx(w,YY1p,'o',w,YY1p,'-',w,YY2p,'x',w,YY2p,'-')
grid
xlabel('Frequency (rad/sec)')
ylabel('Phase deg')
text(1.2,20,'Y1'); text(1.2,182,'Y2')
```

Similarly, to plot the phase angles for $Y_1(j\omega)$ and $Y_2(j\omega)$, use the following commands:

```
Y1p = phase*[1;0];
Y2p = phase*[0;1];
semilogx(w,Y1p,'o',w,Y1p,'-',w,Y2p,'x',w,Y2p,'-')
```

Bode diagrams obtained by use of MATLAB Program 8–16 are shown in Figures 8–103 and 8–104.

A-8-7. Prove that the polar plot of the sinusoidal transfer function

$$G(j\omega) = \frac{j\omega T}{1 + j\omega T}, \qquad \text{for } 0 \le \omega \le \infty$$

is a semicircle. Find the center and radius of the circle.

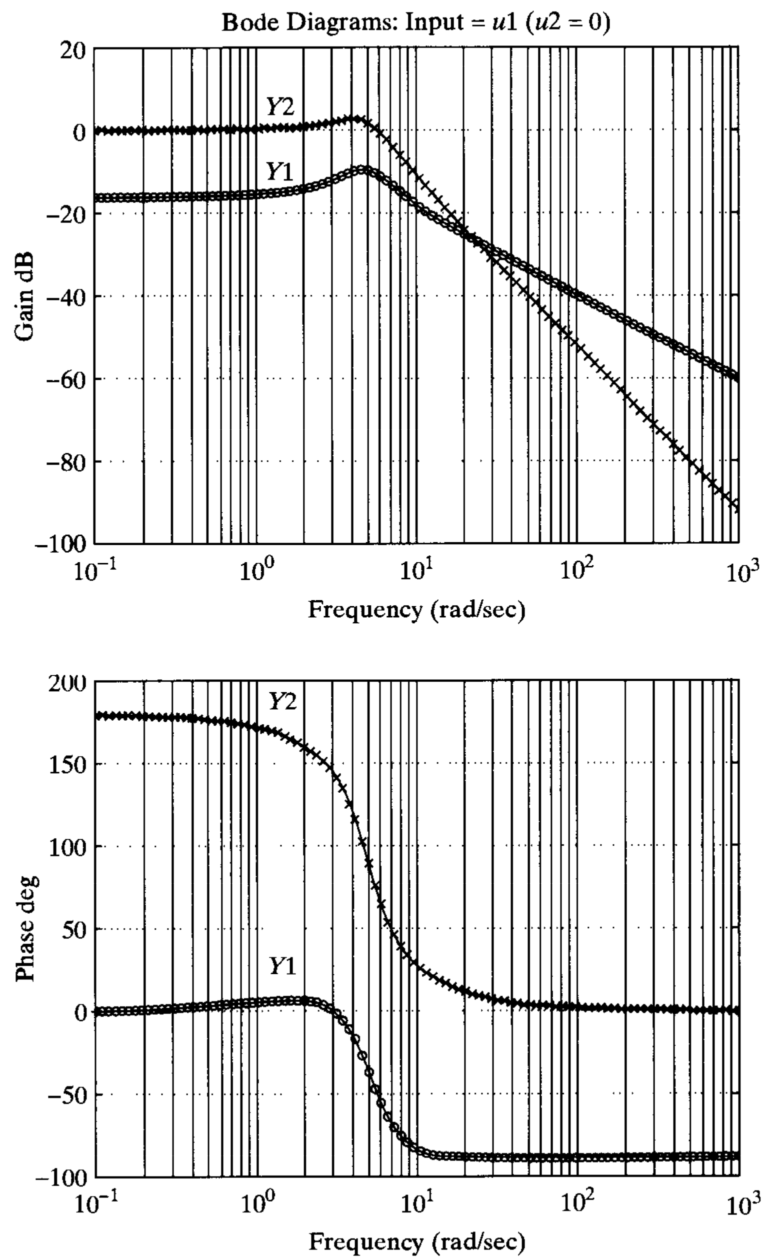

Figure 8–103
Bode diagrams.

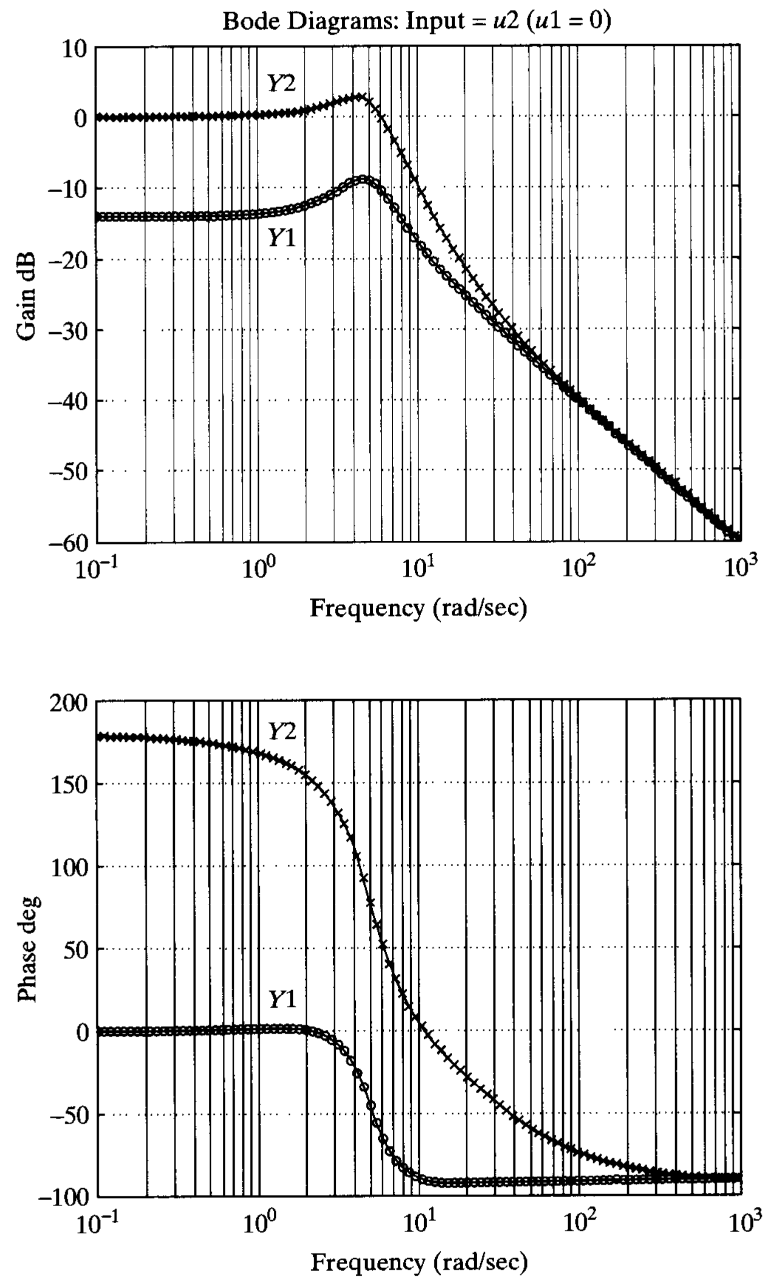

Figure 8–104
Bode diagrams.

Solution. The given sinusoidal transfer function $G(j\omega)$ can be written as follows:

$$G(j\omega) = X + jY$$

where

$$X = \frac{\omega^2 T^2}{1 + \omega^2 T^2}, \qquad Y = \frac{\omega T}{1 + \omega^2 T^2}$$

Then

$$\left(X - \frac{1}{2}\right)^2 + Y^2 = \frac{(\omega^2 T^2 - 1)^2}{4(1 + \omega^2 T^2)^2} + \frac{\omega^2 T^2}{(1 + \omega^2 T^2)^2} = \frac{1}{4}$$

Hence, we see that the plot of $G(j\omega)$ is a circle centered at (0.5,0) with radius equal to 0.5. The upper semicircle corresponds to $0 \leq \omega \leq \infty$, and the lower semicircle corresponds to $-\infty \leq \omega \leq 0$.

A-8-8. Referring to Problem A-8-2, plot the polar locus of $G(s)$ where

$$G(s) = \frac{20(s^2 + s + 0.5)}{s(s + 1)(s + 10)}$$

Locate on the polar locus frequency points where $\omega = 0.1, 0.2, 0.4, 0.6, 1.0, 2.0, 4.0, 6.0, 10.0, 20.0$, and 40.0 rad/sec.

Solution. Noting that

$$G(j\omega) = \frac{2(-\omega^2 + j\omega + 0.5)}{j\omega(j\omega + 1)(0.1\,j\omega + 1)}$$

we have

$$|G(j\omega)| = \frac{2\sqrt{(0.5 - \omega^2)^2 + \omega^2}}{\omega\sqrt{1 + \omega^2}\,\sqrt{1 + 0.01\,\omega^2}}$$

$$\angle G(j\omega) = \tan^{-1}\left(\frac{\omega}{0.5 - \omega^2}\right) - 90^\circ - \tan^{-1}\omega - \tan^{-1}(0.1\omega)$$

The magnitude and phase angle may be obtained as shown in Table 8–3. (Note that the magnitude in decibels and phase angle in degrees can be easily read from Figure 8–98.) The magnitude in decibels can also be easily converted to a number. Figure 8–105 shows the polar plot. Notice the existence of a loop in the polar locus.

Table 8–3 Magnitude and Phase of $G(j\omega)$ Considered in Problem A-8-8

ω	$\lvert G(j\omega)\rvert$	$\angle G(j\omega)$
0.1	9.952	−84.75°
0.2	4.918	−78.96°
0.4	2.435	−64.46°
0.6	1.758	−47.53°
1.0	1.573	−24.15°
2.0	1.768	−14.49°
4.0	1.801	−22.24°
6.0	1.692	−31.10°
10.0	1.407	−45.03°
20.0	0.893	−63.44°
40.0	0.485	−75.96°

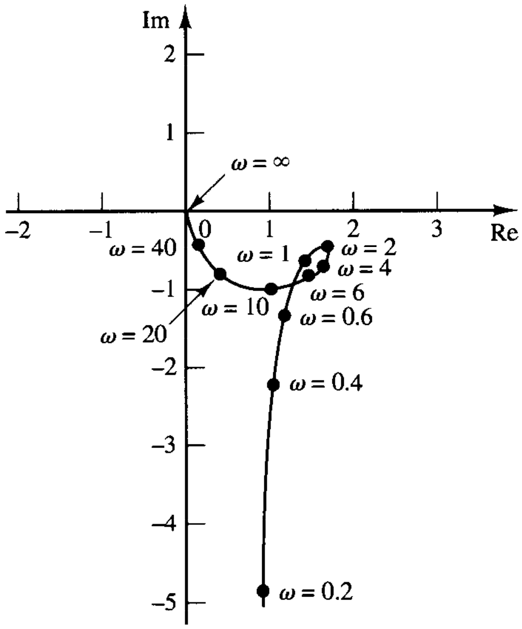

Figure 8–105
Polar plot of
$G(j\omega)$ given in Problem A-8-8.

A-8-9. Consider the function

$$F(s) = \frac{s+1}{s-1}$$

The conformal mapping of the lines $\omega = 0, \pm 1, \pm 2$ and the lines $\sigma = 0, \pm 1, \pm 2$ yield circles in the $F(s)$ plane, as shown in Figure 8–106. Show that if the contour in the s plane encloses the pole of $F(s)$ there is one encirclement of the origin of the $F(s)$ plane in the counterclockwise direction. If the contour in the s plane encloses the zero of $F(s)$, there is one encirclement of the origin of the $F(s)$ plane in the clockwise direction. If the contour in the s plane encloses both the zero and pole or if the contour encloses neither the zero nor pole, then there is no encirclement of the origin of the $F(s)$ plane by the locus of $F(s)$. (Note that in the s plane a representative point s traces out a contour in the clockwise direction.)

Solution. A graphical solution is given in Figure 8–107; this shows closed contours in the s plane and their corresponding closed curves in the $F(s)$ plane.

A-8-10. Prove the following mapping theorem: Let $F(s)$ be a ratio of polynomials in s. Let P be the number of poles and Z be the number of zeros of $F(s)$ that lie inside a closed contour in the s plane, multiplicity accounted for. Let the closed contour be such that it does not pass through any poles or zeros of $F(s)$. The closed contour in the s plane then maps into the $F(s)$ plane as a closed curve. The number N of clockwise encirclements of the origin of the $F(s)$ plane, as a representative point s traces out the entire contour in the s plane in the clockwise direction, is equal to Z–P.

Solution. To prove this theorem, we use Cauchy's theorem and the residue theorem. Cauchy's theorem states that the integral of $F(s)$ around a closed contour in the s plane is zero if $F(s)$ is analytic within and on the closed contour, or

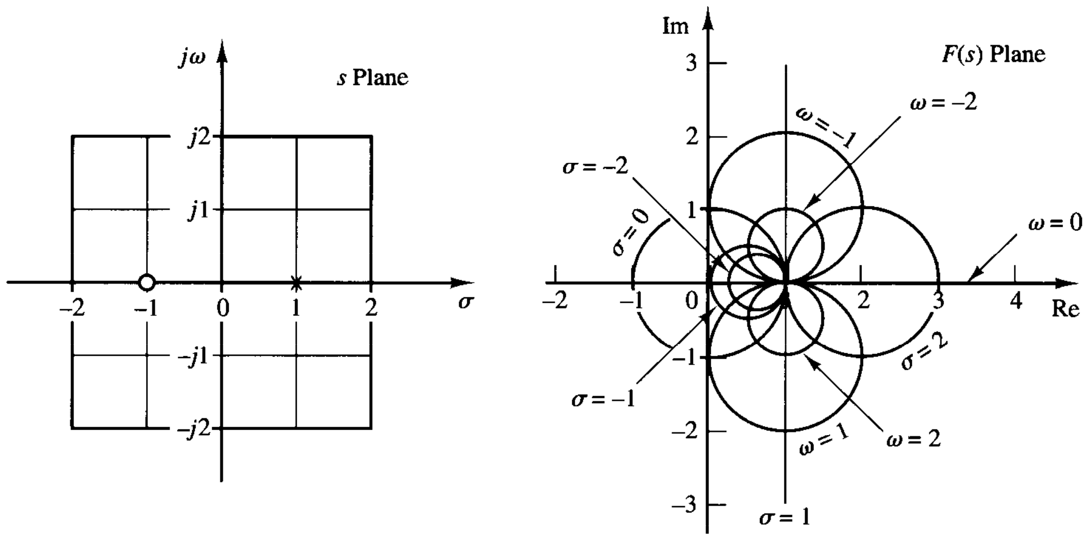

Figure 8–106 Conformal mapping of the s-plane grids into the $F(s)$ plane, where $F(s) = (s + 1)/(s - 1)$.

$$\oint F(s)\, ds = 0$$

Suppose that $F(s)$ is given by

$$F(s) = \frac{(s + z_1)^{k_1}(s + z_2)^{k_2} \cdots}{(s + p_1)^{m_1}(s + p_2)^{m_2} \cdots} X(s)$$

where $X(s)$ is analytic in the closed contour in the s plane and all the poles and zeros are located in the contour. Then the ratio $F'(s)/F(s)$ can be written

$$\frac{F'(s)}{F(s)} = \left(\frac{k_1}{s + z_1} + \frac{k_2}{s + z_2} + \cdots\right) - \left(\frac{m_1}{s + p_1} + \frac{m_2}{s + p_2} + \cdots\right) + \frac{X'(s)}{X(s)} \tag{8–19}$$

This may be seen from the following consideration: If $F(s)$ is given by

$$F(s) = (s + z_1)^k X(s)$$

then $F(s)$ has a zero of kth order at $s = -z_1$. Differentiating $F(s)$ with respect to s yields.

$$F'(s) = k(s + z_1)^{k-1} X(s) + (s + z_1)^k X'(s)$$

Hence,

$$\frac{F'(s)}{F(s)} = \frac{k}{s + z_1} + \frac{X'(s)}{X(s)} \tag{8–20}$$

We see that by taking the ratio $F'(s)/F(s)$ the kth-order zero of $F(s)$ becomes a simple pole of $F'(s)/F(s)$.

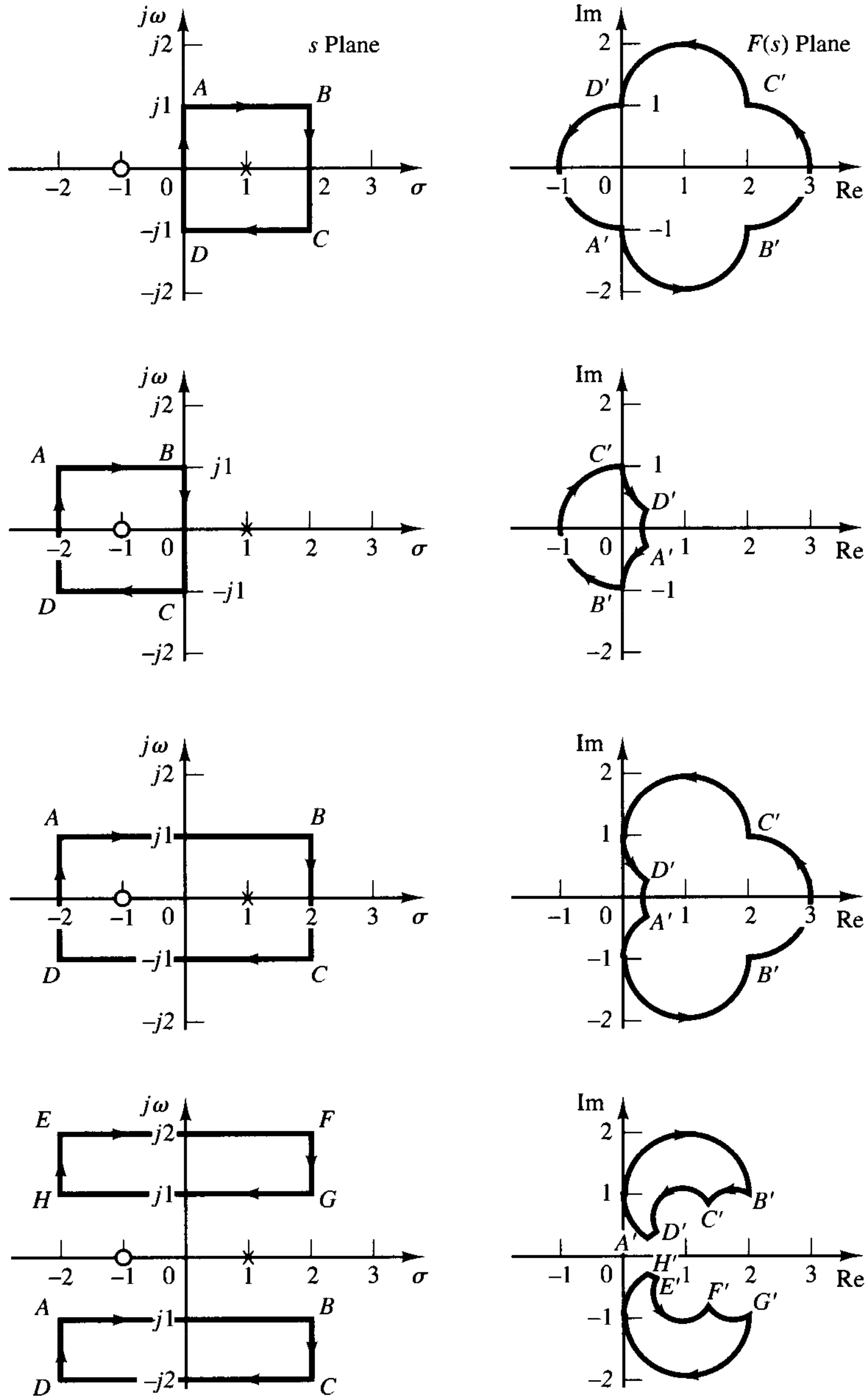

Figure 8–107 Conformal mapping of the s-plane contours into the $F(s)$ plane where $F(s) = (s + 1)/(s - 1)$.

If the last term on the right-hand side of Equation (8–20) does not contain any poles or zeros in the closed contour in the s plane, $F'(s)/F(s)$ is analytic in this contour except at the zero $s = -z_1$. Then, referring to Equation (8–19) and using the residue theorem, which states that the integral of $F'(s)/F(s)$ taken in the clockwise direction around a closed contour in the s plane is equal to $-2\pi j$ times the residues at the simple poles of $F'(s)/F(s)$, or

$$\oint \frac{F'(s)}{F(s)}\, ds = -2\pi j(\Sigma \text{ residues})$$

we have

$$\oint \frac{F'(s)}{F(s)}\,ds = -2\pi j[(k_1 + k_2 + \cdots) - (m_1 + m_2 + \cdots)] = -2\pi j(Z - P)$$

where $Z = k_1 + k_2 + \cdots =$ total number of zeros of $F(s)$ enclosed in the closed contour in the s plane

$P = m_1 + m_2 + \cdots =$ total number of poles of $F(s)$ enclosed in the closed contour in the s plane

[The k multiple zeros (or poles) are considered k zeros (or poles) located at the same point.] Since $F(s)$ is a complex quantity, $F(s)$ can be written

$$F(s) = |F|e^{j\theta}$$

and

$$\ln F(s) = \ln|F| + j\theta$$

Noting that $F'(s)/F(s)$ can be written

$$\frac{F'(s)}{F(s)} = \frac{d\ln F(s)}{ds}$$

we obtain

$$\frac{F'(s)}{F(s)} = \frac{d\ln|F|}{ds} + j\frac{d\theta}{ds}$$

If the closed contour in the s plane is mapped into the closed contour Γ in the $F(s)$ plane, then

$$\oint \frac{F'(s)}{F(s)}\,ds = \oint_\Gamma d\ln|F| + j\oint_\Gamma d\theta = j\int d\theta = 2\pi j(P - Z)$$

The integral $\oint_\Gamma d\ln|F|$ is zero since the magnitude $\ln|F|$ is the same at the initial point and the final point of the contour Γ. Thus we obtain

$$\frac{\theta_2 - \theta_1}{2\pi} = P - Z$$

The angular difference between the final and initial values of θ is equal to the total change in the phase angle of $F'(s)/F(s)$ as a representative point in the s plane moves along the closed contour. Noting that N is the number of clockwise encirclements of the origin of the $F(s)$ plane and $\theta_2 - \theta_1$ is zero or a multiple of 2π rad, we obtain

$$\frac{\theta_2 - \theta_1}{2\pi} = -N$$

Thus, we have the relationship

$$N = Z - P$$

This proves the theorem.

Note that by this mapping theorem the exact numbers of zeros and of poles cannot be found, only their difference. Note also that from Figures 8–108 (a) and (b) we see that, if θ does not change through 2π rad, then the origin of the $F(s)$ plane cannot be encircled.

A-8-11. The Nyquist plot (polar plot) of the open-loop frequency response of a unity-feedback control system is shown in Figure 8–109. Assuming that the Nyquist path in the s plane encloses the entire right-half s plane, draw a complete Nyquist plot in the G plane. Then answer the following questions:

(a) If the open-loop transfer function has no poles in the right-half s plane, is the closed-loop system stable?

(b) If the open-loop transfer function has one pole and no zeros in right-half s plane, is the closed-loop system stable?
(c) If the open-loop transfer function has one zero and no poles in the right-half s plane, is the closed-loop system stable?

Solution. Figure 8–110 shows a complete Nyquist plot in the G plane. The answers to the three questions are as follows:

(a) The closed-loop system is stable, because the critical point $(-1 + j0)$ is not encircled by the Nyquist plot. That is, since $P = 0$ and $N = 0$, we have $Z = N + P = 0$.
(b) The open-loop transfer function has one pole in the right-half s plane. Hence, $P = 1$. (The open-loop system is unstable.) For the closed-loop system to be stable, the Nyquist plot must encircle the critical point $(-1 + j0)$ once counterclockwise. However, the Nyquist plot does not encircle the critical point. Hence, $N = 0$. Therefore, $Z = N + P = 1$. The closed-loop system is unstable.
(c) Since the open-loop transfer function has one zero but no poles in the right-half s plane, we have $Z = N + P = 0$. Thus, the closed-loop system is stable. (Note that the zeros of the open-loop transfer function do not affect the stability of the closed-loop system.)

A-8-12. Is a closed-loop system with the following open-loop transfer function and with $K = 2$ stable?

$$G(s)H(s) = \frac{K}{s(s + 1)(2s + 1)}$$

Find the critical value of the gain K for stability.

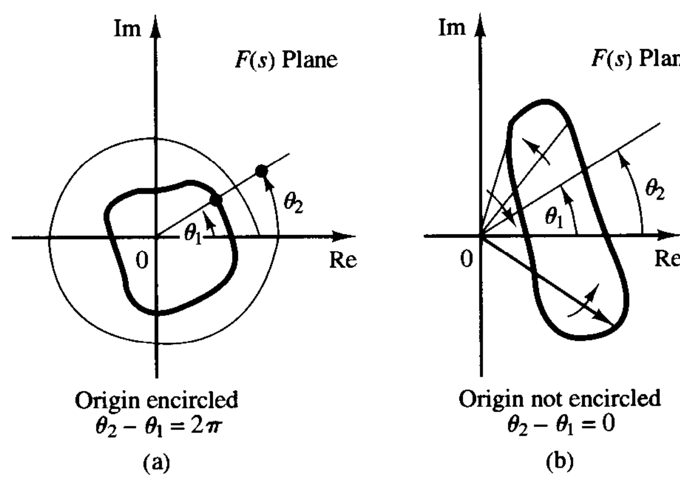

Figure 8–108 Determination of encirclement of the origin of $F(s)$ plane.

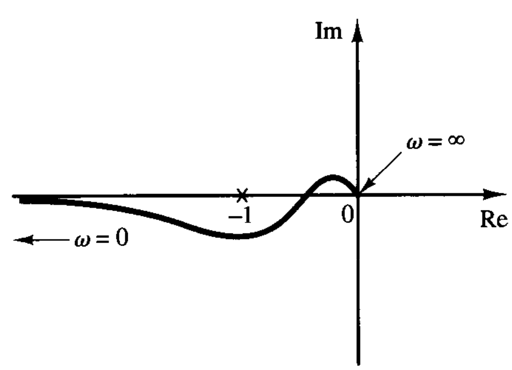

Figure 8–109 Nyquist plot.

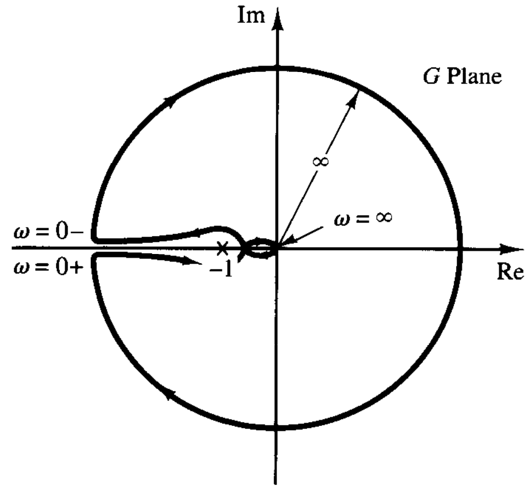

Figure 8–110
Complete Nyquist plot in the G plane.

Solution. The open-loop transfer function is

$$G(j\omega)H(j\omega) = \frac{K}{j\omega(j\omega + 1)(2j\omega + 1)}$$

$$= \frac{K}{-3\omega^2 + j\omega(1 - 2\omega^2)}$$

This open-loop transfer function has no poles in the right-half s plane. Thus, for stability the $-1 + j0$ point should not be encircled by the Nyquist plot. Let us find the point where the Nyquist plot crosses the negative real axis. Let the imaginary part of $G(j\omega)H(j\omega)$ be zero, or

$$1 - 2\omega^2 = 0$$

from which

$$\omega = \pm\frac{1}{\sqrt{2}}$$

Substituting $\omega = 1/\sqrt{2}$ into $G(j\omega)H(j\omega)$, we obtain

$$G\left(j\frac{1}{\sqrt{2}}\right)H\left(j\frac{1}{\sqrt{2}}\right) = -\frac{2K}{3}$$

The critical value of the gain K is obtained by equating $-2K/3$ to -1, or

$$-\frac{2}{3}K = -1$$

Hence,

$$K = \frac{3}{2}$$

The system is stable if $0 < K < \frac{3}{2}$. Hence, the system with $K = 2$ is unstable.

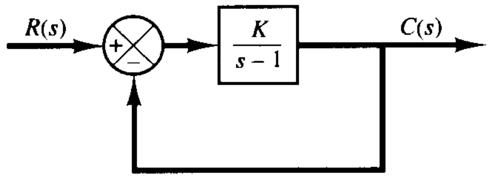

Figure 8–111
Closed-loop system.

A-8-13. Consider the closed-loop system shown in Figure 8–111. Determine the critical value of K for stability by use of the Nyquist stability criterion.

Solution. The polar plot of

$$G(j\omega) = \frac{K}{j\omega - 1}$$

is a circle with center at $-K/2$ on the negative real axis and radius $K/2$, as shown in Figure 8–112 (a). As ω is increased from $-\infty$ to ∞, the $G(j\omega)$ locus makes a counterclockwise rotation. In this system, $P = 1$ because there is one pole of $G(s)$ in the right-half s plane. For the closed-loop system to be stable, Z must be equal to zero. Therefore, $N = Z - P$ must be equal to -1, or there must be one counterclockwise encirclement of the $-1 + j0$ point for stability. (If there is no encirclement of the $-1 + j0$ point, the system is unstable.) Thus, for stability, K must be greater than unity, and $K = 1$ gives the stability limit. Figure 8–112(b) shows both stable and unstable cases of $G(j\omega)$ plots.

A-8-14. Consider a unity-feedback system whose open-loop transfer function is

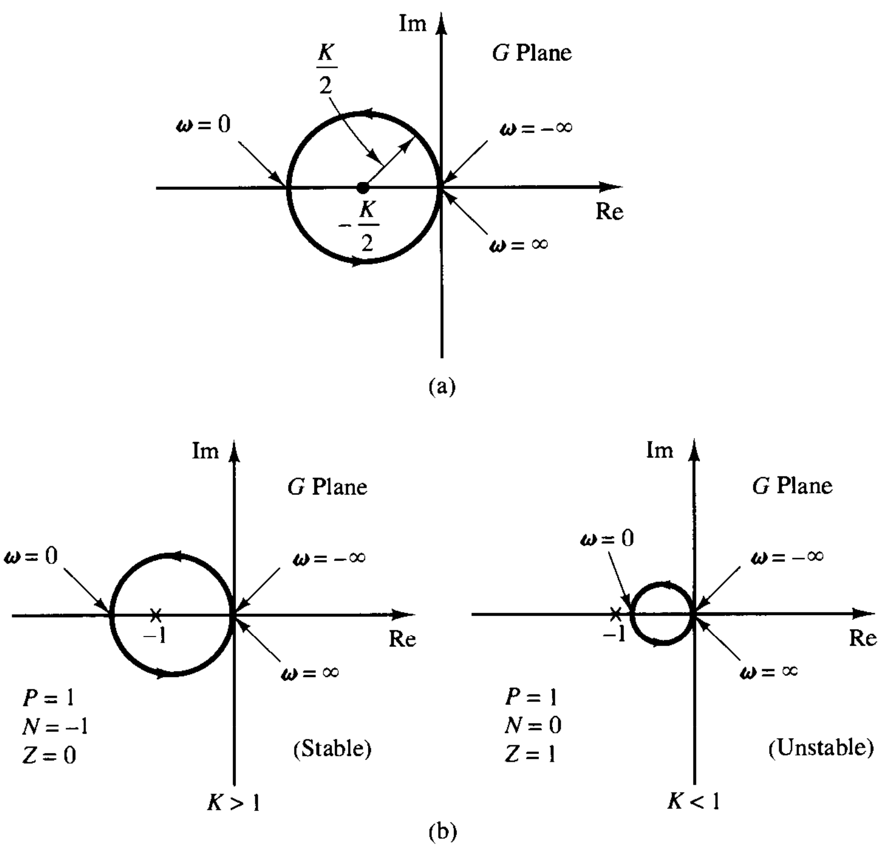

Figure 8–112
(a) Polar plot of $K/(j\omega - 1)$; (b) polar plots of $K/(j\omega - 1)$ for stable and unstable cases.

$$G(s) = \frac{Ke^{-0.8s}}{s+1}$$

Using the Nyquist plot, determine the critical value of K for stability.

Solution. For this system,

$$G(j\omega) = \frac{Ke^{-0.8j\omega}}{j\omega + 1}$$

$$= \frac{K(\cos 0.8\omega - j \sin 0.8\omega)(1 - j\omega)}{1 + \omega^2}$$

$$= \frac{K}{1 + \omega^2} [(\cos 0.8\omega - \omega \sin 0.8\omega) - j(\sin 0.8\omega + \omega \cos 0.8\omega)]$$

The imaginary part of $G(j\omega)$ is equal to zero if

$$\sin 0.8\omega + \omega \cos 0.8\omega = 0$$

Hence,

$$\omega = -\tan 0.8\omega$$

Solving this equation for the smallest positive value of ω, we obtain

$$\omega = 2.4482$$

Substituting $\omega = 2.4482$ into $G(j\omega)$, we obtain

$$G(j2.4482) = \frac{K}{1 + 2.4482^2} (\cos 1.9586 - 2.4482 \sin 1.9586) = -0.378K$$

The critical value of K for stability is obtained by letting $G(j2.4482)$ equal -1. Hence,

$$0.378K = 1$$

or

$$K = 2.65$$

Figure 8–113 shows the Nyquist or polar plots of $2.65e^{-0.8j\omega}/(1 + j\omega)$ and $2.65/(1 + j\omega)$. The first-order system without transport lag is stable for all values of K, but the one with transport lag of 0.8 sec becomes unstable for $K > 2.65$.

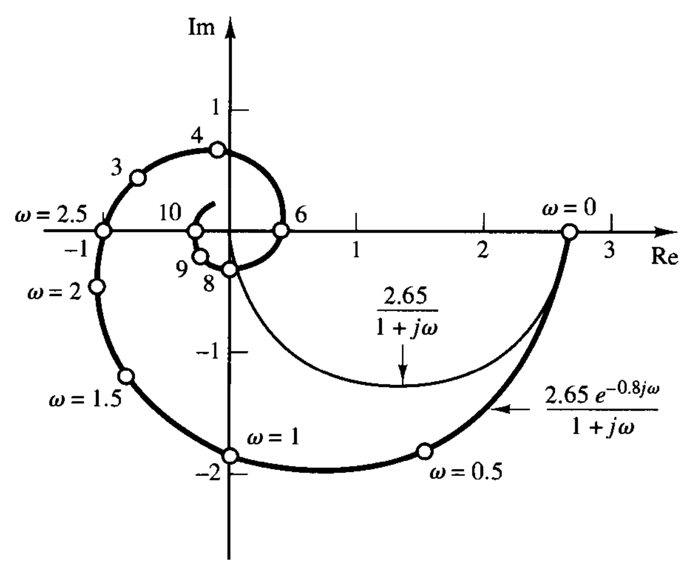

Figure 8–113
Polar plots of $2.65e^{-0.8j\omega}/(1 + j\omega)$ and $2.65/(1 + j\omega)$.

A-8-15. Consider a unity-feedback system with the following open-loop transfer function:

$$G(s) = \frac{20(s^2 + s + 0.5)}{s(s + 1)(s + 10)}$$

Draw a Nyquist plot with MATLAB and examine the stability of the closed-loop system.

Solution. We first enter MATLAB Program 8–17 into the computer. Because in this system MATLAB involves "Divide by zero" in the computation, the resulting Nyquist plot is erroneous, as shown in Figure 8–114.

MATLAB Program 8–17

num = [0 20 20 10];
den = [1 11 10 0];
nyquist(num,den)

This erroneous Nyquist plot can be corrected by entering the axis command, as shown in MATLAB Program 8–18. The resulting Nyquist plot is shown in Figure 8–115.

MATLAB Program 8–18

num = [0 20 20 10];
den = [1 11 10 0];
nyquist(num,d.en)
v = [−2 3 −3 3]; axis(v)
grid
title('Nyquist Plot of G(s) = 20(s^2 + s + 0.5)/[s(s + 1)(s + 10)]')

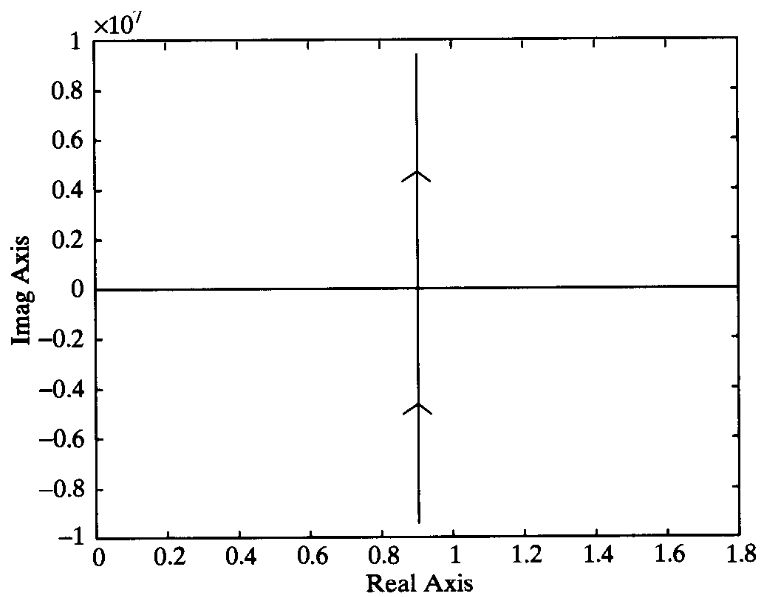

Figure 8–114
Erroneous Nyquist plot.

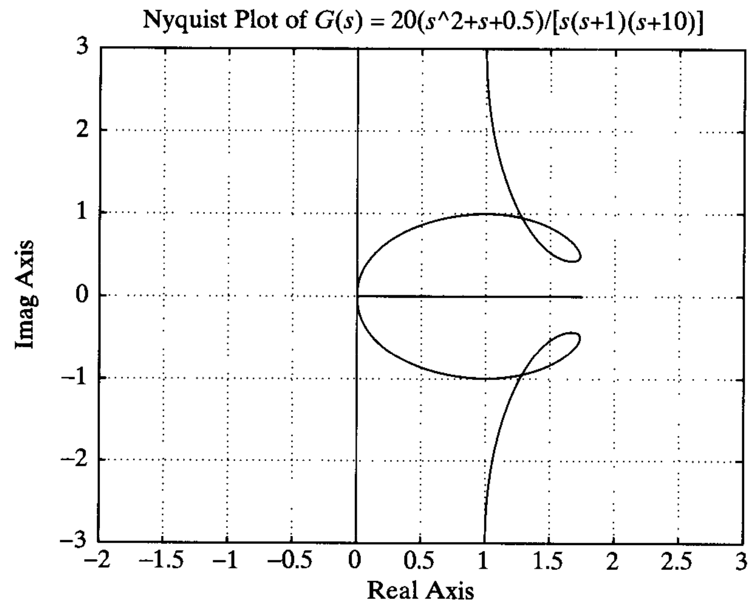

Figure 8–115 Nyquist plot of $G(s) = \dfrac{20(s^2 + s + 0.5)}{s(s+1)(s+10)}$.

Since no open-loop poles lie in the right-half s plane, $P = 0$ in the Nyquist stability criterion. From Figure 8–115 we see that the Nyquist plot does not encircle the $-1 + j0$ point. Hence, the closed-loop system is stable.

A-8-16. Consider the same system as discussed in Problem A-8-15. Draw the Nyquist plot for only the positive frequency region.

Solution. Drawing a Nyquist plot for only the positive frequency region can be done by use of the following command:

```
[re,im,w] = nyquist(num,den,w)
```

The frequency region may be divided into several subregions using different increments. For example, the frequency region of interest may be divided into three subregions as follows:

```
w1 = 0.1:0.1:10;
w2 = 10:2:100;
w3 = 100:10:500;
 w = [w1  w2  w3]
```

MATLAB Program 8–19 uses this frequency region. Using this program, we obtain the Nyquist plot shown in Figure 8–116.

A-8-17. Consider a unity-feedback, positive-feedback system with the following open-loop transfer function:

$$G(s) = \frac{s^2 + 4s + 6}{s^2 + 5s + 4}$$

Draw a Nyquist plot.

Solution. The Nyquist plot of the positive-feedback system can be obtained by defining num and den as

```
num = [-1  -4  -6]
den = [1  5  4]
```

MATLAB Program 8–19

```
% ---------- Nyquist plot ----------

num = [0   20   20   10];
den = [1   11   10   0];
w1 = 0.1:0.1:10; w2 = 10:2:100; w3 =100:10:500;
w = [w1   w2   w3];
[re,im,w] = nyquist(num,den,w);
plot(re,im)
v = [-3   3   -5   1]; axis(v)
grid
title('Nyquist Plot of G(s) = 20(s^2 + s + 0.5)/[s(s + 1)(s + 10)]')
xlabel('Real Axis')
ylabel('Imag Axis')
```

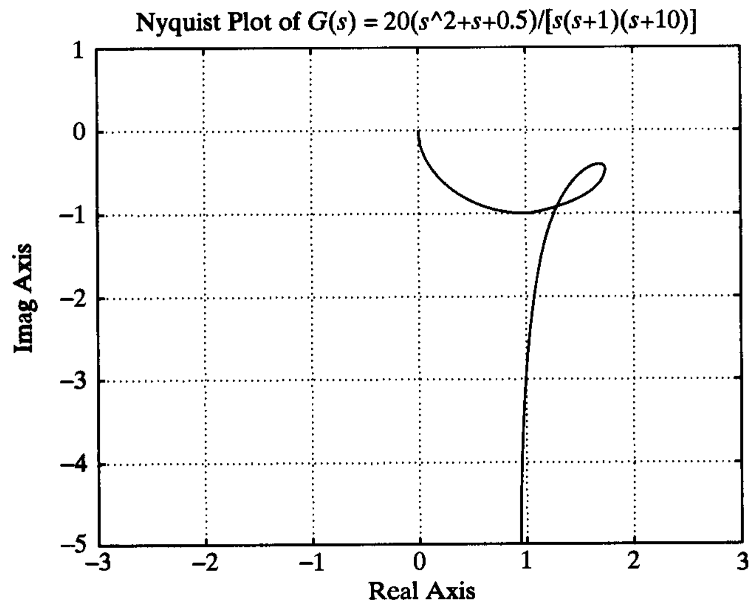

Figure 8–116 Nyquist plot for the positive frequency region.

and using the command nyquist(num,den). MATLAB Program 8–20 produces the Nyquist plot, as shown in Figure 8–117.

MATLAB Program 8–20

```
num = [-1   -4   -6];
den = [1   5   4];
nyquist(num,den);
grid
title('Nyquist Plot of G(s) = -(s^2 + 4s + 6)/(s^2 +5s + 4)')
```

This system is unstable because the $-1 + j0$ point is encircled once clockwise. Note that this is a special case where the Nyquist plot passes through $-1 + j0$ point and also encircles this point once clockwise. This means that the closed-loop system is degenerate; the system behaves as if it

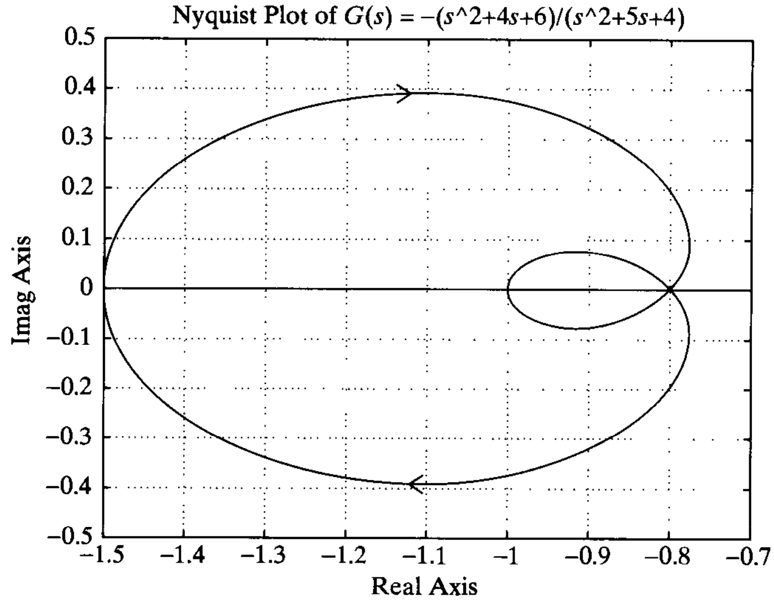

Figure 8–117 Nyquist plot for positive-feedback system.

is an unstable first-order system. See the following closed-loop transfer function of the positive-feedback system:

$$\frac{C(s)}{R(s)} = \frac{s^2 + 4s + 6}{s^2 + 5s + 4 - (s^2 + 4s + 6)}$$

$$= \frac{s^2 + 4s + 6}{s - 2}$$

Note that the Nyquist plot for the positive-feedback case is a mirror image about the imaginary axis of the Nyquist plot for the negative-feedback case. This may be seen from Figure 8–118, which was obtained by use of MATLAB Program 8–21.

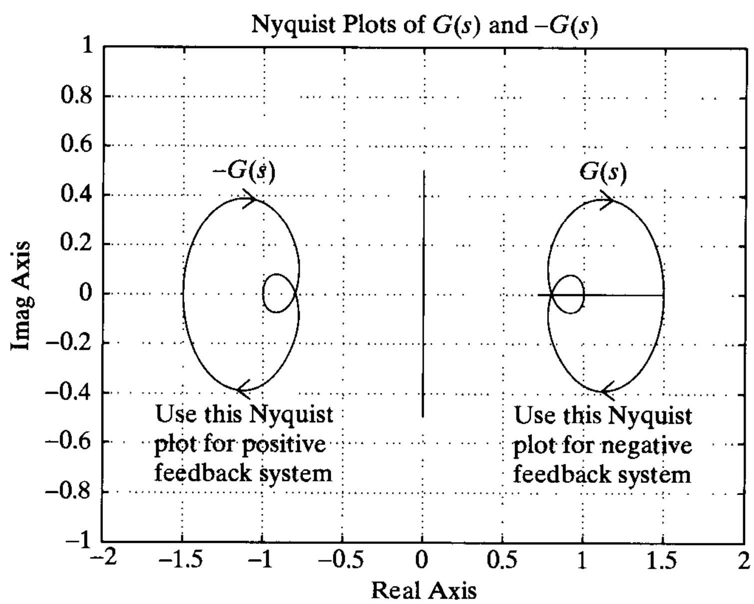

Figure 8–118 Nyquist plots for positive-feedback system and negative-feedback system.

MATLAB Program 8–21

```
num1 = [1  4  6];
den1 = [1  5  4];
num2 = [-1  -4  -6];
den2 = [1  5  4];
nyquist(num1,den1);
hold on
nyquist(num2,den2)
v = [-2  2  -1  1];
axis(v);
grid
title('Nyquist Plots of G(s) and -G(s)')
text(0.95,0.5,'G(s)')
text(0.57,-0.48,'Use this Nyquist')
text(0.57,-0.62,'plot for negative')
text(0.57,-0.73,'feedback system')
text(-1.3,0.5,'-G(s)')
text(-1.7,-0.48,'Use this Nyquist')
text(-1.7,-0.62,'plot for positive')
text(-1.7,-0.73,'feedback system')
```

A-8-18. Suppose that a system possesses at least one pair of complex-conjugate closed-loop poles. If the $-1 + j0$ point is found at the intersection of a constant-σ curve and a constant-ω curve in the $G(s)$ plane, then those particular values of σ and ω, which we define as $-\sigma_c$ and ω_c, respectively, characterize the closed-loop pole closest to the $j\omega$ axis in the upper-half s plane. (Note that $-\sigma_c$ represents the exponential decay and ω_c represents the damped natural frequency of the step transient-response term due to the pair of the closed-loop poles closest to the $j\omega$ axis.) Probable values of $-\sigma_c$ and ω_c may be estimated from the plot, as shown in Figure 8–119. Thus, the pair of complex-conjugate closed-loop poles that lies closest to the $j\omega$ axis can be determined graphically. It should be noted that all closed-loop poles are mapped into the $-1 + j0$ point in the $G(s)$ plane. Although the complex-conjugate closed-loop poles closest to the $j\omega$ axis can be found easily by this technique, finding other closed-loop poles, if any, by this technique is practically impossible.

If the data on $G(j\omega)$ are experimental, then a curvilinear square near the $-1 + j0$ point can be constructed by extrapolation. Referring to Figure 8–120, we can find the location of the dominant closed-loop poles in the s plane, or the damping ratio ζ and the damped natural frequency ω_d, by drawing the line AB that connects the $-1 + j0$ point (point A) and point B, the nearest approach to the $-1 + j0$ point, and then constructing a curvilinear square $CDEF$, as shown in

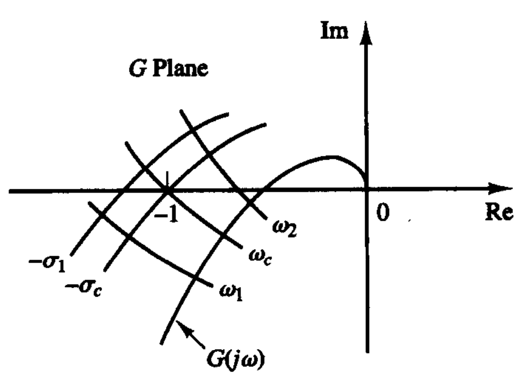

Figure 8–119
Estimation of $-\sigma_c$ and ω_c.

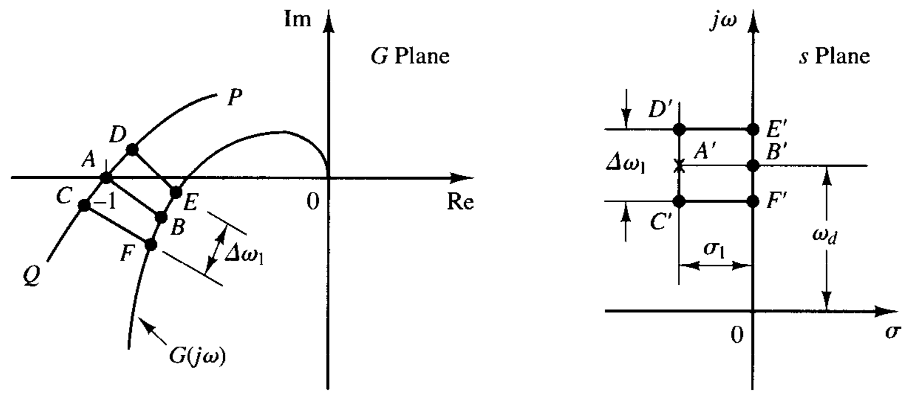

Figure 8–120 Conformal mapping of a curvilinear square near the $-1 + j0$ point in the $G(s)$ plane into the s plane.

Figure 8–120. This curvilinear square $CDEF$ may be constructed by drawing the most likely curve PQ (where PQ is the conformal mapping of a line parallel to the $j\omega$ axis in the s plane) passing through the $-1 + j0$ point and "parallel" to the $G(j\omega)$ locus, and adjusting the points $C, D, E,$ and F such that $\overset{\frown}{FB} = \overset{\frown}{BE}$, $\overset{\frown}{CA} = \overset{\frown}{AD}$, and $\overset{\frown}{FE} + \overset{\frown}{CD} = \overset{\frown}{FC} + \overset{\frown}{ED}$. The corresponding s-plane contour $C'D'E'F'$, together with point A', the closed-loop pole closest to the $j\omega$ axis, is shown in Figure 8–120. The value of the frequency interval $\Delta\omega_1$ between points E and F is approximately equal to the value of σ_1 shown in Figure 8–120. The frequency at point B is approximately equal to the damped natural frequency ω_d. The closed-loop poles closest to the $j\omega$ axis are then estimated as

$$s = -\sigma_1 \pm j\omega_d$$

Then the damping ratio ζ of these closed-loop poles can be obtained from

$$\frac{\zeta}{\sqrt{1-\zeta^2}} = \frac{\sigma_1}{\omega_d} = \frac{\Delta\omega_1}{\omega_d}$$

It should be noted that the damped natural frequency ω_d of the step transient response actually is on the frequency contour that passes through the $-1 + j0$ point and is not necessarily the point of nearest approach to the $G(j\omega)$ locus. Therefore, the value ω_d obtained by the technique above is somewhat in error.

From our analysis, we may conclude that it is possible to estimate the closed-loop poles closest to the $j\omega$ axis from the closeness of approach of the $G(j\omega)$ locus to the $-1 + j0$ point, the frequency at the point of nearest approach, and the frequency graduation near this point.

Referring to the frequency-response plot of $G(j\omega)$ of a unity-feedback system, as shown in Figure 8–121, find the closed-loop poles closest to the $j\omega$ axis.

Solution. The line connecting the $-1 + j0$ point and the point of nearest approach of the $G(j\omega)$ locus to the $-1 + j0$ point is drawn first. Then the curvilinear square $ABCD$ is constructed. Since the frequency at the point of nearest approach is $\omega = 2.9$, the damped natural frequency is approximately 2.9, or $\omega_d = 2.9$. From the curvilinear square $ABCD$, it is found that

$$\Delta\omega = \omega_D - \omega_A = 3.4 - 2.4 = 1.0$$

The closed-loop poles closest to the $j\omega$ axis are then estimated as

$$s = -1 \pm j2.9$$

The $G(j\omega)$ locus shown in Figure 8–121 is actually a plot of the following open-loop transfer function:

$$G(s) = \frac{5(s+20)}{s(s+4.59)(s^2+3.41s+16.35)}$$

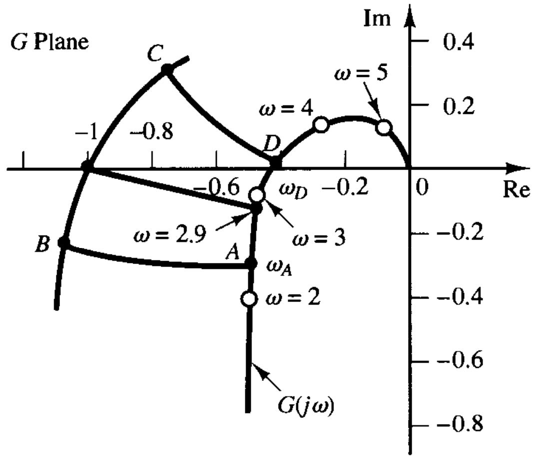

Figure 8–121
Polar plot and a curvilinear square.

The exact closed-loop poles of this system are $s = -1 \pm j3$ and $s = -3 \pm j1$. The closed loop poles closest to the $j\omega$ axis are $s = -1 \pm j3$. In this particular example, we see that the error involved is fairly small. In general, this error depends on a particular $G(j\omega)$ curve. The nearer the $G(j\omega)$ locus is to the $-1 + j0$ point, the smaller the error.

A-8-19. Figure 8–122 shows a block diagram of a space vehicle control system. Determine the gain K such that the phase margin is 50°. What is the gain margin in this case?

Solution. Since

$$G(j\omega) = \frac{K(j\omega + 2)}{(j\omega)^2}$$

we have

$$\underline{/G(j\omega)} = \underline{/j\omega + 2} - 2\,\underline{/j\omega} = \tan^{-1}\frac{\omega}{2} - 180°$$

The requirement that the phase margin be 50° means that $\underline{/G(j\omega_c)}$ must be equal to $-130°$, where ω_c is the gain crossover frequency, or

$$\underline{/G(j\omega_c)} = -130°$$

Hence, we set

$$\tan^{-1}\frac{\omega_c}{2} = 50°$$

from which we obtain

$$\omega_c = 2.3835 \text{ rad/sec}$$

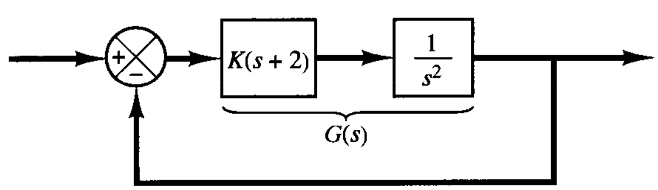

Figure 8–122
Space vehicle control system.

Since the phase curve never crosses the $-180°$ line, the gain margin is $+\infty$ dB. Noting that the magnitude of $G(j\omega)$ must be equal to 0 dB at $\omega = 2.3835$, we have

$$\left| \frac{K(j\omega + 2)}{(j\omega)^2} \right|_{\omega = 2.3835} = 1$$

from which we get

$$K = \frac{2.3835^2}{\sqrt{2^2 + 2.3835^2}} = 1.8259$$

This K value will give the phase margin of 50°.

A-8-20. Draw a Bode diagram of the open-loop transfer function $G(s)$ of the closed-loop system shown in Figure 8–123. Determine the phase margin and gain margin.

Solution. Note that

$$G(j\omega) = \frac{20(j\omega + 1)}{j\omega(j\omega + 5)[(j\omega)^2 + 2j\omega + 10]}$$

$$= \frac{0.4(j\omega + 1)}{j\omega(0.2j\omega + 1)\left[\left(\frac{j\omega}{\sqrt{10}}\right)^2 + \frac{2}{10}j\omega + 1\right]}$$

The quadratic term in the denominator has the corner frequency of $\sqrt{10}$ rad/sec and the damping ratio ζ of 0.3162, or

$$\omega_n = \sqrt{10}, \qquad \zeta = 0.3162$$

The Bode diagram of $G(j\omega)$ is shown in Figure 8–124. From this diagram we find the phase margin to be 100° and the gain margin to be +13.3 dB.

A-8-21. For the standard second-order system

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

show that the bandwidth ω_b is given by

$$\omega_b = \omega_n(1 - 2\zeta^2 + \sqrt{4\zeta^4 - 4\zeta^2 + 2})^{1/2}$$

Note that ω_b/ω_n is a function only of ζ. Plot a curve ω_b/ω_n versus ζ.

Solution. The bandwidth ω_b is determined from $|C(j\omega_b)/R(j\omega_b)| = -3$ dB. Quite often, instead of -3 dB, we use -3.01 dB, which is equal to 0.707. Thus,

$$\left| \frac{C(j\omega_b)}{R(j\omega_b)} \right| = \left| \frac{\omega_n^2}{(j\omega_b)^2 + 2\zeta\omega_n(j\omega_b) + \omega_n^2} \right| = 0.707$$

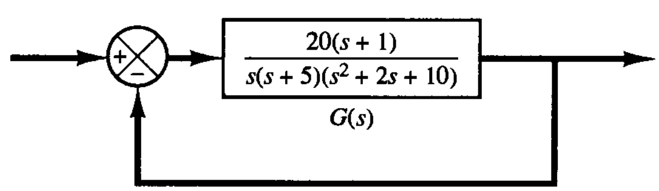

Figure 8–123
Closed-loop system.

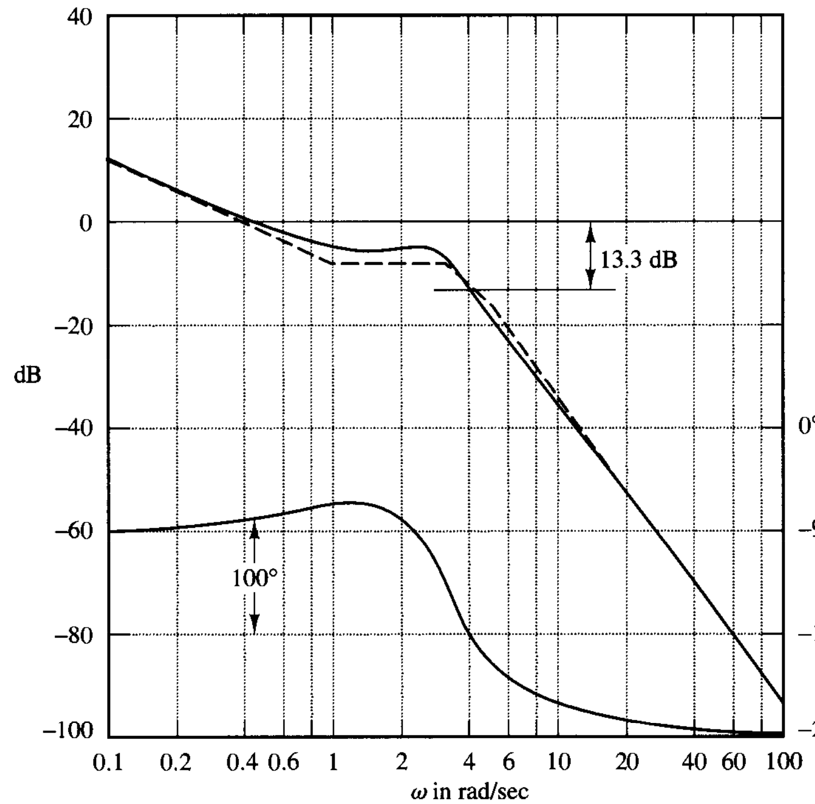

Figure 8–124 Bode diagram of $G(j\omega)$ of system shown in Figure 8–123.

Then

$$\frac{\omega_n^2}{\sqrt{(\omega_n^2 - \omega_b^2)^2 + (2\zeta\omega_n\omega_b)^2}} = 0.707$$

from which we get

$$\omega_n^4 = 0.5[(\omega_n^2 - \omega_b^2)^2 + 4\zeta^2\omega_n^2\omega_b^2]$$

By dividing both sides of this last equation by ω_n^4, we obtain

$$1 = 0.5\left\{\left[1 - \left(\frac{\omega_b}{\omega_n}\right)^2\right]^2 + 4\zeta^2\left(\frac{\omega_b}{\omega_n}\right)^2\right\}$$

Solving this last equation for $(\omega_b/\omega_n)^2$ yields

$$\left(\frac{\omega_b}{\omega_n}\right)^2 = -2\zeta^2 + 1 \pm \sqrt{4\zeta^4 - 4\zeta^2 + 2}$$

Since $(\omega_b/\omega_n)^2 > 0$, we take the plus sign in this last equation. Then

$$\omega_b^2 = \omega_n^2(1 - 2\zeta^2 + \sqrt{4\zeta^4 - 4\zeta^2 + 2})$$

or

$$\omega_b = \omega_n(1 - 2\zeta^2 + \sqrt{4\zeta^4 - 4\zeta^2 + 2})^{1/2}$$

Figure 8–125 shows a curve relating ω_b/ω_n versus ζ.

A-8-22. A unity-feedback control system has the following open-loop transfer function:

$$G(s) = \frac{K}{s(s+1)(s+2)}$$